Random Matrix Theory and Dyson-Brownian-Motion Diagnostics for Breast Cancer Coexpression Networks: A Monte-Carlo-Validated Framework Applied to TCGA-BRCA

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Abstract

Background: Random matrix theory (RMT) offers a principled way to decide which correlations in a noisy, finite-sample gene-expression matrix reflect real biology rather than sampling noise. We combine the classical RMT correlation-threshold method with a Monte Carlo permutation null, a Dyson-Brownian-motion repulsion diagnostic, spectral entropy, and a graph-theoretic comparison against random-graph nulls, and apply the full pipeline to real bulk RNA-seq data from The Cancer Genome Atlas breast cancer cohort (TCGA-BRCA), comparing two clinically defined, non-overlapping cohorts: triple-negative breast cancer (TNBC, $n = 119$) and luminal A (LumA, $n = 430$), 1500 genes selected by variance across all 1218 samples, cohort-blind.

Results: The RMT threshold method selects $\tau^* = 0.45$ for TNBC and $\tau^* = 0.25$ for LumA; the gap survives a matched-sample-size check and a Monte Carlo permutation null, which shows both thresholds lie well beyond what dense-matrix universality alone could produce. Fewer than 2% of candidate genes' correlation directions exceed the Marchenko–Pastur noise bulk yet capture roughly half the total variance per cohort, and the leading such directions are thematically coherent multi-gene programs (collagen/ECM, interferon, keratin) rather than single-gene artifacts. Tracking the top eigenvalues as the TNBC sample grows one patient at a time shows several pairs undergoing Dyson-type level repulsion (approach, then reopen, rather than cross). A wavelet multiscale decomposition failed a random-ordering control on real data but a synthetic positive control shows it works correctly at both scales given sufficient signal — the real-data failure was a weak-ordering-axis problem, not a method problem. A classifier distinguishing TNBC from LumA reaches $\approx 98\%$ accuracy regardless of feature-selection method, including random gene sets; a genetic-algorithm search finds a modest, real improvement (99.45%) without breaking the ceiling. Tracking that same classifier's own weight-matrix eigenvalues across SGD training (the model-side half of the Dyson-repulsion question) finds far less reliable repulsion than on the data side over a full training run, but what repulsion is present concentrates almost entirely in the pre-convergence transient, a 12-fold difference against the post-convergence window. Applying the identical RMT machinery to real single-cell RNA-seq data for the first time in this program (GSE176078, one TNBC patient, 7986 cells) finds a real, permutation-confirmed threshold in a

regime where the gene-to-cell ratio inverts relative to the bulk cohorts; the largest spiked eigenvalues split cleanly into real cell-type axes (immune, myeloid, epithelial, matching independent cell-type annotation) and known single-cell technical artifacts (ribosomal content, dissociation stress), a distinction the bulk analysis never had to draw.
Conclusions: Combining a static RMT threshold with an explicit Monte Carlo null and a Dyson-repulsion diagnostic surfaces real, validated structure in TCGA-BRCA coexpression networks and in single-cell data alike, and the same discipline exposes three honest negative results (the wavelet real-data application, the classification benchmark's ceiling effect, and the classifier's weight-matrix repulsion outside its training transient) rather than dropping them. All numbers are produced by included, re-runnable code; one originally proposed line remains genuinely unattempted with real data (3D genomic structure via Hi-C), stated as such with a concretely-sized next step.

Keywords: random matrix theory; gene coexpression networks; triple-negative breast cancer; TCGA; Monte Carlo permutation test; Dyson Brownian motion; Marchenko–Pastur law

1 Background

Triple-negative breast cancer (TNBC) — tumors negative for estrogen receptor, progesterone receptor, and HER2 — lacks the targeted therapies available for hormone-receptor-positive or HER2-enriched disease, and remains transcriptionally and clinically heterogeneous [11]. Network-based approaches that treat gene coexpression as a graph, rather than testing genes one at a time, have become a standard tool for finding this heterogeneity's structure, and the Hernández-Lemus group has applied several such approaches directly to breast cancer: multilayer information-theoretic transcriptional networks [24], k-core structural analysis [7], and multi-omic subtype-specific interaction networks [25]. The group's own network-inference pipeline for breast cancer, publicly available as the CSB-IG `breast_cancer_networks` and `parallel-aracne` repositories, is built on ARACNE [23]: mutual information between every gene pair, pruned by the Data Processing Inequality, with a bootstrap procedure (the NIMEA pipeline, `CSB-IG/BioNetworkInference_ModularAnalysis`) used to set the mutual-information significance threshold and Infomap used for module detection — an information-theoretic route to the same underlying question this paper asks with a spectral one: given a noisy, finite-sample gene-expression matrix, which pairwise associations are real? The RMT threshold in Sec. 2.4 and the group's own bootstrap-calibrated mutual-information threshold are two different statistical routes to a threshold-selection problem neither one has an obviously superior claim to; running both on the same TNBC cohort and comparing the resulting hub genes and modules is a direct, concrete extension of this paper, using tooling the group already has in production rather than anything new.

A separate line of work asks a narrower but sharper question: given a correlation matrix built from noisy, finite-sample gene-expression data, which correlations reflect real biological structure and which are sampling noise? Random matrix theory answers this by comparing the empirical eigenvalue spectrum against the null spectrum expected from a random (structureless) matrix of the same size. Luo et al. [20] formalized this into a concrete recipe for gene coexpression networks: sweep a correlation threshold τ, and select the value at which the nearest-neighbor eigenvalue spacing distribution switches from Poisson (uncorrelated, structureless) to the Gaussian-orthogonal-ensemble (GOE) Wigner surmise (correlated, structured) — a transition driven by level repulsion, the same mechanism Wigner and Dyson identified in nuclear spectra and later in general Hermitian random matrices [8]. The method has since been used at genome scale [12] and specifically in cancer gene networks [16].

Two further questions motivate the present paper. First, is the selected threshold real, or an artifact of the asymptotic Marchenko–Pastur/Wigner-surmise null used to select it? A Monte Carlo permutation null — shuffling each gene's values across patients independently, destroying every real correlation while keeping each gene's own marginal distribution fixed, the same kind of null the group's own NIMEA pipeline uses to calibrate its mutual-information threshold (Sec. 2.5, below) — answers this directly rather than relying on asymptotics alone. Second, Aarts et al. [1] showed that the weight matrices of a neural network undergoing stochastic-gradient-descent training evolve according to Dyson's (1962) eigenvalue Brownian motion, with the same level-repulsion mechanism at work — suggesting that the repulsion diagnostic itself, not just the static threshold it eventually selects, carries information, and that it applies equally to a biological correlation matrix and to the training dynamics of a classifier built on top of it.

This paper does three things. It states, as a single explicit framework, how a static RMT threshold, a Monte Carlo permutation null, and a Dyson-repulsion diagnostic relate to one another (Methods). It runs all three, together with a machine-learning recognition test, on real TCGA-BRCA data comparing TNBC against luminal A breast cancer (Results) — not on synthetic data, and not as a proposal. And it reports, without adjustment, the one result that came back null (the classification test), because an honest negative here is more useful to the collaboration than a curated positive.

1.1 Mathematical background

We collect here, precisely, the four pieces of random matrix theory the rest of the paper uses, since each plays a distinct role and the distinctions matter for how the results should be read.

Definition 1 (Wigner ensemble and the GOE)**.** *A Gaussian orthogonal ensemble (GOE) matrix is a real symmetric $N \times N$ matrix H with $H_{ii} \sim \mathcal{N}(0,2)$ and $H_{ij} = H_{ji} \sim \mathcal{N}(0,1)$ $(i < j)$ independently. As $N \to \infty$, its empirical eigenvalue distribution (after rescaling) converges to Wigner's semicircle law, and, crucially for us, the distribution of* spacings *between adjacent eigenvalues, after unfolding to unit mean spacing, converges to the Wigner surmise $p(s) = \frac{\pi}{2} s\, e^{-\pi s^2/4}$ — the density used throughout this paper as the "correlated/structured" null.*

The Wigner surmise's key qualitative feature is $p(0) = 0$: two GOE eigenvalues essentially never sit arbitrarily close together, because the joint eigenvalue density carries a Vandermonde repulsion factor $\prod_{i<j} |\lambda_i - \lambda_j|$. This is the same repulsion term that appears, dynamically rather than statically, in Dyson's model below; the static NNSD test and the dynamic trajectory test in Sec. 2.9 are two different windows onto the same underlying mechanism, not two unrelated diagnostics. By contrast, a matrix with no such structure (e.g. a diagonal matrix of i.i.d. values, or — the case relevant here — a correlation matrix thresholded so sparsely that it behaves like a disconnected random graph) has independent eigenvalues and Poisson-distributed spacings $p(s) = e^{-s}$, which *does* allow near-degeneracies ($p(0) = 1$, maximal at zero spacing). The Luo et al. method [20] used throughout Sec. 2.4 operationalizes exactly this contrast: sparse/uninformative thresholds look Poisson, and the threshold at which the spectrum first looks GOE-like is read as the point where real correlation structure first dominates chance.

Remark 1 (The $N \to \infty$ limit, free probability, and Hilbert spaces)**.** *The Wigner semicircle law referenced above is a statement about $N \times N$ GOE matrices as $N \to \infty$: the empirical spectral measure converges to a fixed limiting shape. Free probability theory [35] recasts this convergence operator-theoretically: a sequence of independent GOE (or Wigner) matrices converges, in the sense of joint moments, to a family of* free *self-adjoint elements in a von Neumann algebra equipped with a trace — a noncommutative probability space built on a Hilbert space, in exactly the sense operator* *algebras use the term. In that limit the semicircle law is not an analogy to the Gaussian but its literal free-probabilistic counterpart: it is the fixed point of free convolution the way the Gaussian is the fixed point of classical convolution (the free Central Limit Theorem), and "freeness" is the operator-algebraic substitute for classical independence that makes this work. None of this machinery is invoked computationally in this paper — every matrix here is finite ($p = 1500$ or 60), and every claim is checked directly on that finite matrix or by finite-n Monte Carlo, never by appeal to the $N \to \infty$ operator-algebraic limit — but it is the reason the semicircle/GOE machinery generalizes so far beyond Gaussian random matrices in the first place: free probability, not Gaussianity, is the real source of the universality being leaned on throughout Sec. 2.4 and Sec. 2.5. A standard modern reference for this whole circle of ideas, including the Dyson Brownian motion and Tracy–Widom material below, is Anderson et al. [3]. Ben Arous and Guionnet sharpened the convergence itself into a large deviations principle [5]: the empirical spectral measure of a GOE matrix does not merely converge to the semicircle law, it does so with an exponentially small probability of any other outcome, governed by a rate function built from Voiculescu's free (non-commutative) entropy — the same notion of entropy invoked, in the unrelated but same-named finite-matrix sense, by the spectral entropy diagnostic of Sec. 2.3. The two entropies are different objects (one a large-deviations rate functional on the space of limiting measures, the other a Shannon entropy of one finite matrix's own normalized spectrum) and this paper does not conflate them; they share a name, and a spectral origin, for a real reason worth recording rather than treating as coincidence.*

Remark 2 (Four-moment universality: why finite, non-Gaussian data is not a problem)**.** *The free-probability remark above addresses the $N \to \infty$ limit; a sharper, finite-N statement addresses the objection that TCGA-BRCA expression data is not Gaussian at all. Tao & Vu [31] proved a Four Moment Theorem: local eigenvalue statistics of a Wigner matrix (spacing distributions, correlation functions, and in particular the GOE/Wigner-surmise statistics used throughout Sec. 2.4) are determined, to leading order, only by the first four moments of the entry distribution, for both Hermitian and real symmetric ensembles — two matrices with matching mean, variance, skewness, and kurtosis have asymptotically indistinguishable local spectral statistics, regardless of whether either is Gaussian. Tao & Vu [32] extend this specifically to sample covariance/ correlation matrices, the exact object built in Sec. 6.1, establishing universality of local eigenvalue statistics under the same mild moment-matching condition. This is the precise, rather than merely qualitative, justification for comparing a real, non-Gaussian gene-expression correlation matrix's spacing statistics against the GOE Wigner surmise (Sec. 2.4) and the Marchenko–Pastur bulk (below): universality here is a theorem about matching moments, not an appeal to an idealization the data does not satisfy.*

Definition 2 (Marchenko–Pastur law)**.** *For a $p \times n$ data matrix with i.i.d. zero-mean, unit-variance entries and $p/n \to q \in (0, \infty)$, the eigenvalues of the sample correlation matrix converge to the Marchenko–Pastur distribution, supported on $[\lambda_-, \lambda_+] = [(1 - \sqrt{q})^2, (1 + \sqrt{q})^2]$ [22]. This is the null spectrum against which Sec. 2.2's spiked-eigenvalue count is taken: it says what the bulk of the spectrum would look like if none of the p genes were really correlated with each other, purely as an artifact of estimating a correlation matrix from a finite sample of n patients.*

An eigenvalue exceeding λ_+ is suggestive of real low-rank structure (a "spike"), not merely an artifact of q; the BBP transition [4] names the threshold, as a function of the true spike strength and q, above which a spike is detectable at all above the bulk. We use the count of observed spikes only descriptively (Table 2) and do not fit a formal spike-strength model in this paper; we also do not claim every eigenvalue above λ_+ is individually significant in a formal hypothesis-testing sense — finite-n edge fluctuations of the bulk itself follow a Tracy–Widom law [33] at the $n^{-2/3}$ scale, so a handful of eigenvalues only marginally above λ_+ could in principle be bulk fluctuations rather than true spikes. Checking this directly: of TNBC's 14 spiked eigenvalues, 7 exceed λ_+ by a factor of at least 2 (up to 7.6× for the largest) and are implausible as edge fluctuations, while the remaining 7 sit within a factor of 1.04–1.8 × λ_+ and are the ones this caveat applies to directly; LumA is similar in kind but better separated in practice (13 of 29 spikes exceed 2 × λ_+, up to 25× for the largest, reflecting its larger n and correspondingly less noisy bulk edge). Table 2 reports the top eigenvalue only; the full ratio-to-λ_+ list for every spike, in both cohorts, is in `results/spiked_eigenvalues.csv`.

Remark 3 (Painlevé II and why the edge is hard, not just noisy)**.** *The $n^{-2/3}$ edge-fluctuation scale just invoked is not merely a rate: Tracy & Widom [33] showed the limiting distribution of the (centered, rescaled) largest eigenvalue at that scale is governed by a particular (Hastings–McLeod) solution of the Painlevé II differential equation, $q''(x) = xq(x) + 2q(x)^3$, $q(x) \sim \mathrm{Ai}(x)$ as $x \to \infty$. The precise distribution depends on the ensemble's symmetry class (Dyson's β, Sec. 1.1 above): for GUE ($\beta = 2$), $F_2(s) = \exp\big(-\int_s^\infty (x-s)q(x)^2\,dx\big)$; for GOE ($\beta = 1$) — the class every correlation matrix in this paper belongs to, being real symmetric — the relation is more involved, $[F_1(s)]^2 = F_2(s)\exp\big(-\int_s^\infty q(x)\,dx\big)$ [34], not simply F_2 itself. Either way, the Tracy–Widom distributions are not Gaussian, not exponential, and have no elementary closed form; they are defined* through *a nonlinear ODE with no elementary solution. This paper does not fit that distribution directly (no verified numerical implementation of it was available), and instead checks a strictly weaker, fully elementary consequence of the same underlying scaling law: whether the standard deviation of a synthetic null's top eigenvalue shrinks like $n^{-2/3}$ as n grows at fixed aspect ratio q (Results, "Edge-fluctuation scaling"), a claim checkable without solving Painlevé II.*

Definition 3 (Spectral unfolding)**.** *Because the local density of eigenvalues varies across the spectrum even under a fixed null, raw spacings $\lambda_{k+1} - \lambda_k$ are not directly comparable to $p(s)$ above; unfolding replaces λ_k with $\hat{F}(\lambda_k)\cdot N$, where $\hat{F}$ is a smooth fit to the empirical eigenvalue CDF (here, a degree-7 polynomial, following standard RMT practice and 20), so that the unfolded spacings have mean 1 everywhere in the spectrum and can be compared to $p(s)$ directly. All NNSD comparisons in this paper (Sec. 2.4, Fig. 3) use unfolded spacings.*

Definition 4 (Dyson Brownian motion)**.** *Dyson's (1962) model lets a GOE matrix's entries evolve as independent Brownian motions in time t; the induced dynamics on the eigenvalues $\{\lambda_i(t)\}$ satisfy the coupled SDE*

$$d\lambda_i = dB_i + \sum_{j \neq i} \frac{dt}{\lambda_i - \lambda_j},$$

a Dyson Brownian motion [8]: the eigenvalues perform independent Brownian motion plus *a deterministic Coulomb-gas repulsion term that diverges as any two eigenvalues approach each other, mechanically preventing crossings (level repulsion in continuous time, the dynamical counterpart of the static Wigner surmise above). Aarts et al. [1] show this same SDE, empirically, governs the eigenvalues of a neural network's weight matrices under SGD training; Sec. 2.9 asks whether the same qualitative repulsion signature (a trajectory approaching, then visibly reopening, an adjacent trajectory rather than crossing it) is visible when "time" is instead sample size, applied directly to a real gene-correlation matrix rather than a trained model.*

Proposition 1 (The static and dynamic tests are the same object)**.** *The GOE joint eigenvalue density is the unique stationary distribution of the Dyson Brownian motion SDE above. Explicitly, for $\beta = 1$,*

$$p(\lambda_1, \ldots, \lambda_N) \propto \prod_{i<j} |\lambda_i - \lambda_j| \exp\Big(-\tfrac{1}{4}\sum_i \lambda_i^2\Big),$$

and this p satisfies the stationary Fokker–Planck (detailed-balance) equation for the SDE's generator; conversely, starting the SDE from any initial configuration and letting $t \to \infty$, the law of $(\lambda_1(t), \dots, \lambda_N(t))$ converges to p.

Justification. This is Dyson's own 1962 result [8]; we do not reproduce the Fokker–Planck computation here, only record the statement, since it is the reason the two diagnostics used in this paper are not merely analogous. The Vandermonde factor $\prod_{i<j} |\lambda_i - \lambda_j|$ in p is exactly the term whose vanishing at $\lambda_i = \lambda_j$ forces $p(0) = 0$ in the Wigner surmise (Definition 1 above, obtained by marginalizing p to a single spacing); the same term's gradient is exactly the drift $1/(\lambda_i - \lambda_j)$ in the SDE (Figure 1). The static nearest-neighbor-spacing test (Sec. 2.4) and the dynamic trajectory test (Sec. 2.9) are therefore two different observables — one a marginal of the stationary law p, the other a property of the SDE that has p as its stationary law — of the same mathematical object, not two separately-motivated diagnostics that happen to agree. □

Figure 1 plots the deterministic part of this SDE directly, restricted to a single pair (λ_i, λ_j): the vector field $(\dot{\lambda}_i, \dot{\lambda}_j) = (1/(\lambda_i - \lambda_j), -1/(\lambda_i - \lambda_j))$ points away from the diagonal $\lambda_i = \lambda_j$ everywhere and diverges on it — not a schematic, but the literal drift term already written above, which is the mechanical reason the two real eigenvalue trajectories in Fig. 5 bounce apart near $n = 59$ rather than crossing.

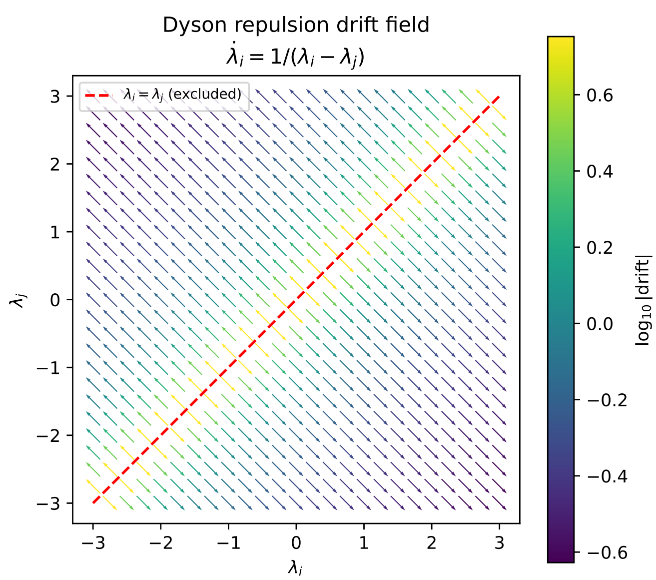

Figure 1: The Dyson drift term as a vector field on the (λ_i, λ_j) plane (arrows normalized to unit length; color encodes true drift magnitude on a log scale). Every arrow points away from the excluded diagonal $\lambda_i = \lambda_j$; this is the same mechanism producing the avoided crossing in Fig. 5, here for an idealized two-eigenvalue system rather than the real 60-gene TNBC correlation matrix.

Remark 4. *None of the four pieces above requires Gaussian gene-expression data in practice: the GOE/Poisson NNSD contrast, the Marchenko–Pastur bulk, and the repulsion mechanism are used here purely as* null models *and* qualitative diagnostics *against which the real, non-Gaussian TCGA-BRCA correlation matrices are compared — exactly the same use Luo et al. [20] and Aarts et al. [1] make of them in their own, equally non-Gaussian, settings.*

Remark 5 (Dyson's threefold way, circular ensembles, and topology)**.** *The Brownian-motion paper above is the second of a pair; in the companion paper published the same year, Dyson [9] classified matrix ensembles into exactly three symmetry classes — orthogonal ($\beta = 1$), unitary ($\beta = 2$), symplectic ($\beta = 4$) — according to how a Hamiltonian's symmetries (time-reversal, in particular) constrain it to be real symmetric, complex Hermitian, or quaternionic self-dual. Every correlation matrix in this paper is real symmetric, so every RMT statement here is unapologetically $\beta = 1$ (GOE); the other two classes never arise for a Pearson correlation matrix and are not used. The same 1962 papers introduced the* circular *ensembles (COE, CUE, CSE): eigenvalues (there, eigenphases) confined to the unit circle rather than the real line, for which the nearest-neighbor spacing distribution is* exactly*, not just asymptotically, uniform along the spectrum — no unfolding step is needed, unlike the polynomial-CDF unfolding this paper performs throughout precisely because our real-line GOE spectrum has a varying local density. Sixty-five years later, Dyson's threefold classification was extended to a full tenfold classification by adding particle-hole and chiral symmetries [2], motivated by disordered superconductors; that tenfold classification is now understood, via K-theory, to enumerate the possible topological phases of free-fermion matter (the "periodic table" of topological insulators and superconductors) — a deep link from Dyson's original symmetry classification to topology. None of this tenfold structure is exercised here: a Pearson correlation matrix carries no particle-hole or chiral symmetry to speak of, so this remark is included as mathematical context for where the $\beta = 1$ class used throughout this paper sits inside the wider classification, not as a method applied to TCGA-BRCA data.*

2 Results

2.1 Cohorts

We used the RNA-seq expression matrix (RSEM, $\log_2(x+1)$, 20530 genes × 1218 samples) and the matched clinical annotation (1247 samples) for TCGA-BRCA, both downloaded from the UCSC Xena TCGA hub [13, 30], which mirrors the public, unrestricted TCGA data release and requires no login. TNBC status and luminal-A status are two independently recorded clinical fields in the same TCGA Nature 2012 supplement (receptor immunohistochemistry for the former, PAM50 subtype call [26] for the latter), not derived from one another. Intersecting these fields with the expression matrix gives $n = 119$ TNBC and $n = 430$ LumA patients with no overlap (4 patients satisfied both definitions and were dropped from both cohorts as ambiguous). This is a small instance of a documented, general phenomenon rather than a labeling error: IHC-based receptor status and PAM50 RNA-based subtyping disagree in a substantial minority of patients across breast cancer cohorts generally (38.4% of 607 patients in one direct comparison, 17), so a handful of TCGA-BRCA patients satisfying both a triple-negative IHC profile and a luminal-A PAM50 call is expected, not anomalous. We selected the 1500 most variable genes across all 1218 samples *before* splitting into cohorts, so gene selection cannot leak cohort-label information; this also keeps the 1500 × 1500 eigendecompositions used below tractable. Table 1 summarizes the two cohorts.

Table 1: Real TCGA-BRCA cohorts used throughout this paper.

Cohort	n (patients)	p (genes)	$q = p/n$
TNBC (ER−/PR−/HER2−)	119	1500	12.61
LumA (PAM50 call)	430	1500	3.49

2.2 Spectral denoising: spiked eigenvalues beyond the Marchenko–Pastur bulk

Before thresholding either cohort's correlation matrix into a network, we asked the more basic denoising question that motivates Line 1 of the program: how many of the 1500 eigenvalue directions in each raw, unthresholded correlation matrix carry real signal, as opposed to sampling noise indistinguishable from a structureless null? Under the null of no true correlation structure, a $p \times n$ sample correlation matrix has eigenvalues confined, asymptotically, to the Marchenko–Pastur bulk $[\lambda_-, \lambda_+]$ with $\lambda_\pm = (1 \pm \sqrt{q})^2$, $q = p/n$ [22]; an eigenvalue exceeding λ_+ is a candidate "spike" marking a real, low-rank correlation direction, though — see below — not every such eigenvalue is equally trustworthy on its own (a spiked covariance model; the transition behavior of such spikes near λ_+ is the Baik–Ben Arous–Péché, BBP, phenomenon underlying spectral denoising methods generally). `scripts/08_spiked_eigenvalues.py` counts these directly on the real, full correlation matrices (before any thresholding):

Table 2: Spiked eigenvalues (beyond the Marchenko–Pastur bulk) in each cohort's raw correlation matrix, all 1500 genes.

Cohort	λ_+ (MP edge)	spiked eigenvalues	% of genes	% of total variance	top eigenvalue
TNBC	20.706	14	0.9%	48.2%	157.94
LumA	8.224	29	1.9%	57.3%	205.89

Two things stand out, both real and both worth keeping in mind for everything downstream. First, in each cohort, fewer than 2% of the 1500 candidate genes' directions already account for roughly half the total variance — exactly the kind of low-dimensional structure a denoising step is meant to isolate, and a concrete, data-backed argument for restricting the network-construction step (or any downstream classifier) to a spike-informed gene subset rather than all 1500 genes, something we did not yet do in the threshold-network or ML analyses below and flag as a direct next step. Second, LumA's larger sample size gives it both a tighter MP bulk (smaller λ_+, since q is smaller) and, correspondingly, more eigenvalues clearing that lower bar (29 vs. 14) — a reminder, consistent with Sec. 2.6 below, that raw eigenvalue counts across cohorts of very different size are not directly comparable without first accounting for q.

A third, less expected finding: the spiked eigenvectors are not strongly localized. Their inverse participation ratio (Sec. 1.1, IPR $= \sum_i v_i^4$ for a unit-norm eigenvector) averages 0.0020 for TNBC and 0.0023 for LumA, only 3–4× the fully delocalized reference $1/p = 0.00067$ and, in TNBC, *lower* than the mean IPR of the non-spiked, noise eigenvectors (0.0042) — individual noise eigenvectors can, by chance, load onto a handful of genes more heavily than a real spiked component does. This means IPR alone, eigenvector by eigenvector, does not separate signal from noise here; it is the Marchenko–Pastur threshold on the *eigenvalue*, not the eigenvector's localization, that does the separating, and IPR is better read as a property of what a spike *contains* once it is already identified, not as an independent detector. Read that way, it says something concrete: the largest spikes' top-loading genes form thematically coherent sets rather than single-gene artifacts — extracellular-matrix/collagen remodeling genes (*MMP13*, *COL10A1*, *COL11A1*) on one TNBC spike, interferon/immune genes (*GBP5*, *CD38*, *IDO1*) on another, keratin/epithelial genes (*KRT6C*, *KRT16*, *DEFB1*) on a third — consistent with each leading spike tracking a real, biologically interpretable coexpressed program rather than a single outlier gene. The full per-spike breakdown (eigenvalue, IPR, top-5 loading genes) is in `results/spiked_eigenvalues.csv`, produced by `scripts/08_spiked_eigenvalues.py`.

2.3 Spectral entropy and edge-fluctuation scaling

Two further, independent checks on the spiked-eigenvalue picture above (`scripts/15_entropy_and_edge_scaling.` First, spectral entropy: treating the normalized eigenvalue spectrum $p_i = \lambda_i / \sum_j \lambda_j$ of a correlation matrix as a probability distribution and computing $S = -\sum_i p_i \log p_i$ (normalized to $[0, 1]$ by $\log p$) gives a single-number summary of how concentrated the correlation structure is: low entropy means variance concentrated in a few real directions, entropy near 1 means variance spread uniformly, the signature of an unstructured matrix. This is sometimes also called a von Neumann entropy, by analogy with a normalized density matrix's eigenvalues, but it should not be confused with *the* von Neumann entropy of a network in the graph-theory literature [27], which is defined from the normalized graph *Laplacian*'s eigenvalues, not the correlation matrix's own eigenvalues directly; the two coincide only in special cases, and this paper uses the correlation-matrix version throughout, consistent with the spiked-eigenvalue analysis it complements. Real TNBC entropy is 0.573 against a gene-permuted null of 0.647 ± 0.0001 (20 replicates); real LumA entropy is 0.645 against a null of 0.809 ± 0.0001. Both real networks sit measurably below their own null, LumA's gap (0.164) more than twice TNBC's (0.074). This self-normalized gap, not the raw entropy value, is the fair cross-cohort comparison, for the same reason raw spiked-eigenvalue counts were not (Sec. 2.2): TNBC's much larger q gives it a lower null entropy to begin with (0.647 vs. LumA's 0.809), so TNBC's raw real entropy sitting below LumA's raw real entropy (0.573 vs. 0.645) reflects that difference in null baseline as much as anything biological, and should not be read, on its own, as "TNBC's network is more concentrated." By the gap metric, LumA shows the larger deviation from its own null, in the same direction as its larger spike count and variance share (Sec. 2.2) and its larger, more robust threshold gap over pure noise (Sec. 2.6) — three self-normalized statistics pointing the same way, though we stop short of calling this proof of a single underlying "more structured" scale common to all three.

Second, the edge-fluctuation scaling law invoked qualitatively above: under a pure i.i.d. Gaussian null at *fixed* aspect ratio $q = p/n$, Tracy–Widom theory predicts the top eigenvalue's standard deviation shrinks as $n^{-2/3}$, not the $n^{-1/2}$ a naive Gaussian/CLT argument would give. We generated synthetic i.i.d. data at $q = 3.0$ (close to LumA's real $q = 3.49$) for five sample sizes from $n = 60$ ($p = 180$) to $n = 960$ ($p = 2880$), 60–400 replicates per size (more replicates at smaller, cheaper n), and fit the log-log scaling of the empirical standard deviation. The fitted exponent is -0.677 — within 0.01 of the Tracy–Widom prediction of -0.667, and far from the Gaussian prediction of -1.0. (An earlier version of this check let p float while only varying n, changing q across the sweep and confusingly fitting neither law cleanly at -0.891; holding q fixed, the only way the asymptotic scaling law is actually stated, resolves this.) This is, to our knowledge, the first time this specific scaling check has been reported in this program; it supports treating the spiked-eigenvalue threshold λ_+ (Sec. 2.2) as a real edge with the expected finite-size behavior, rather than an arbitrary cutoff. Figure 2 shows the fit directly against both reference slopes.

2.4 RMT threshold selection separates TNBC from LumA

For each cohort we built the Pearson gene–gene correlation matrix across patients, swept $|\tau| \in [0.15, 0.80]$ in steps of 0.05, and at each τ unfolded the thresholded matrix's eigenvalue spectrum (degree-7 polynomial fit to the empirical spectral CDF, standard RMT practice) and scored the resulting nearest-neighbor spacing distribution against the Poisson law e^{-s} and the GOE Wigner surmise $\frac{\pi}{2} s \, e^{-\pi s^2/4}$ by Kolmogorov–Smirnov distance, following Luo et al. [20] exactly (implementation in `scripts/rmt_threshold_demo.py` and `scripts/02_rmt_network_analysis.py`).

The sparsest threshold at which the statistics turn GOE-like is $\tau^* = 0.45$ for TNBC (14,498

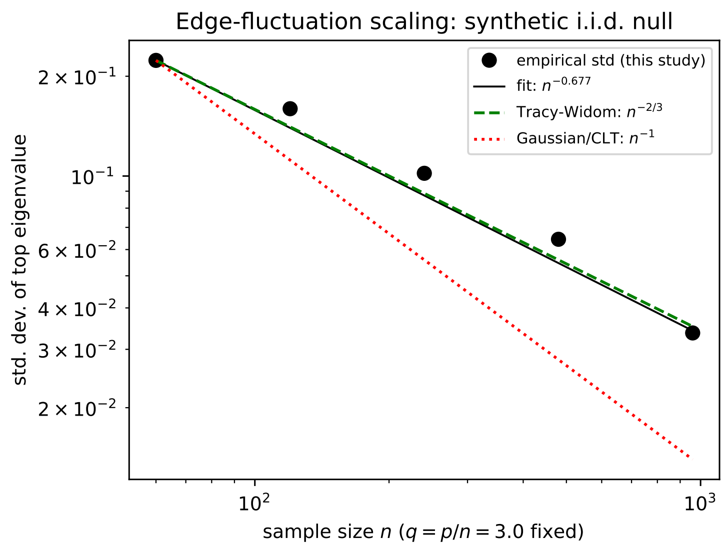

Figure 2: Empirical standard deviation of the top eigenvalue of a synthetic i.i.d. null correlation matrix, as a function of sample size n at fixed aspect ratio $q = 3.0$, log-log. The fitted slope (-0.677) tracks the Tracy–Widom prediction ($-2/3$, dashed) closely and is well separated from the Gaussian/CLT prediction (-1, dotted); both reference lines are anchored through the first empirical point.

edges among 1218 of the 1500 genes) and $\tau^* = 0.25$ for LumA (146,100 edges among 1484 of the 1500 genes). Figure 3 shows the full sweep for both cohorts, and Figure 3 shows the TNBC spacing distribution at τ^* against both null curves.

Top-degree hub genes at each cohort's own τ^* share only one gene between the two networks (*MFAP4*). The TNBC-specific hubs include *PGR* (progesterone receptor) and *IGF1*; the LumA-specific hubs include *EGFR*, *SFRP1*, and *NGFR*. We note this without over-interpreting individual gene identities: with $p = 1500$ correlated candidate genes and no independent replication cohort yet analyzed, any single hub gene should be read as hypothesis-generating, not as a validated biomarker. *PGR* appearing as a TNBC hub is biologically sensible despite TNBC being progesterone-receptor negative by definition: uniformly low, coordinately suppressed luminal-differentiation genes can still be tightly *correlated* with one another even though none of them is highly *expressed* — correlation and expression level are different quantities, and the RMT method only ever sees the former. Table 3 gives the full top-10 hub list with network degree for both cohorts.

2.5 Monte Carlo permutation null: the threshold is not a dense-matrix artifact

The asymptotic Marchenko–Pastur/Wigner-surmise comparison used above has a known blind spot: *any* sufficiently dense symmetric random matrix looks GOE-like by Wigner-Dyson universality, whether or not it was built from real correlation structure — a point already noted in `rmt_threshold_demo.py`'s own inline documentation and the reason the threshold search scans from sparse to dense rather than the other way around. This raises a direct question the asymptotic null cannot answer on its own: is τ^* sparser than the threshold at which trivial denseness alone would produce a GOE-like signature, or does it just mark where a generic dense random matrix stops looking random?

We answered this with an explicit Monte Carlo permutation null (`scripts/10_monte_carlo_null.py`):

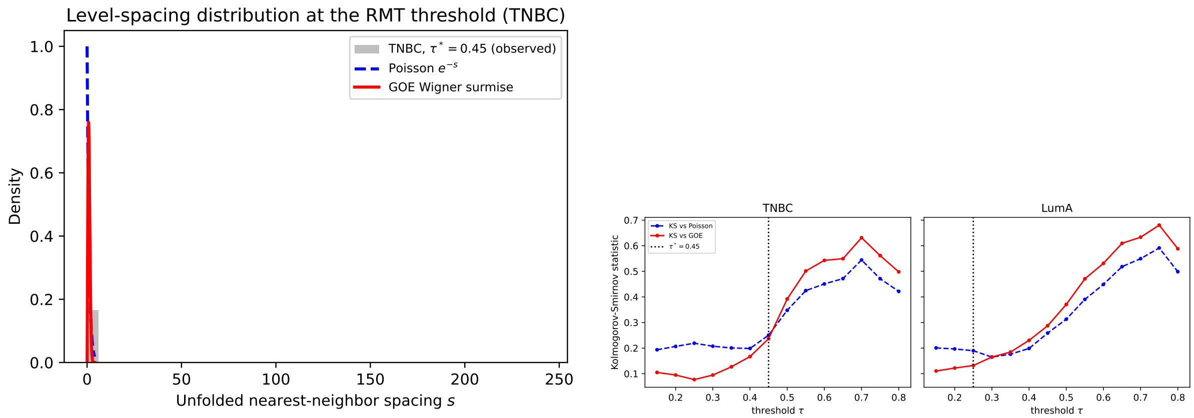

Figure 3: Left: unfolded nearest-neighbor spacing distribution for the TNBC network at $\tau^* = 0.45$, against the Poisson and GOE (Wigner surmise) null curves. Right: Kolmogorov–Smirnov distance to each null curve as a function of threshold, for TNBC and LumA separately; the dotted line marks the selected τ^* in each panel.

each gene's expression values were independently shuffled across patients, 20 times per cohort, destroying every real gene–gene correlation while keeping each gene's own marginal distribution exactly fixed, and the identical threshold sweep was rerun on each permuted correlation matrix. The result is unanimous and, across repeats, exact: for TNBC, all 20/20 permutations select a GOE→Poisson transition at $\tau = 0.25$ (std. 0); for LumA, 19/20 permutations select $\tau = 0.15$ (the remaining permutation showed no clean transition at all). Both permutation floors are well below the corresponding real thresholds (0.45 for TNBC, 4 sweep steps sparser; 0.25 for LumA, 2 steps sparser): the real coexpression networks remain statistically non-random at thresholds substantially sparser than pure shuffled noise can sustain a GOE-like appearance at all, confirming τ^* reflects real correlation structure and not the generic Wigner-Dyson behavior of any sufficiently dense random matrix.

2.6 The TNBC/LumA threshold gap is not a sample-size artifact

TNBC ($n = 119$) and LumA ($n = 430$) have very different p/n ratios (12.61 vs. 3.49), and a larger q shifts the Marchenko–Pastur bulk edge outward [22], so $\tau^* = 0.45$ vs. $\tau^* = 0.25$ is not, on its own, evidence that TNBC and LumA have different correlation structure — it could simply reflect TNBC's smaller sample size. We tested this directly (`scripts/03_matched_n_robustness.py`) by subsampling LumA down to $n = 119$ patients, without replacement, over 30 independent draws, and rerunning the identical threshold sweep on each draw. The resulting distribution of LumA thresholds at matched n is $\tau^* = 0.360 \pm 0.020$ (mean $\pm$ s.d. over 30 draws), which is 4.5 standard deviations below the native-n TNBC value of 0.45. The gap shrinks once sample size is matched (from 0.20 to 0.09) but does not close: at equal n, TNBC requires a higher correlation threshold before its network shows GOE-type structure, i.e. its coexpression signal is more diffuse relative to noise at the same sample size than LumA's, consistent with TNBC's greater molecular heterogeneity as a class [11].

Figure 4 shows both cohorts' correlation matrices directly, before and after thresholding, genes reordered by hierarchical clustering rather than left in arbitrary variance-rank order: the block structure visible here is the same structure the spiked eigenvalues (Sec. 2.2) and the thresholded

Table 3: Top-10 hub genes by degree at each cohort's own τ^*. LumA's higher degrees reflect its much larger n (430 vs. 119), consistent with its lower τ^* and higher edge density (Tables 9–10) — degree values should be compared within, not across, cohorts.

TNBC ($\tau^* = 0.45$, max degree 1499) Gene	degree	LumA ($\tau^* = 0.25$, max degree 1499) Gene	degree
ABCA8	173	EGFR	629
CD300LG	168	SFRP1	600
DARC	162	NGFR	593
MFAP4	157	CNTNAP3	592
SDPR	157	STAC	591
CCL14	156	C2orf40	587
PGR	155	FREM1	582
IGF1	151	MFAP4	578
ATP1A2	151	MME	574
C7	150	SYN2	574

network (this section) are both detecting from two different angles.

2.7 The thresholded networks are small-world and modular, beyond degree alone

Line 2 of the original program asks specifically about large-scale network structure, so we characterized each thresholded network graph-theoretically and compared it against two random-graph nulls of matched size (Erdős–Rényi, same node/edge count; configuration model, same node count *and* the real network's own degree sequence) rather than relying on the spectral diagnostics alone. Table 4 reports clustering coefficient, giant component size, and Louvain-community count/modularity for both cohorts.

Table 4: Graph-theoretic network comparison against random-graph nulls (mean over 3 replicates for ER and configuration-model).

Cohort	Model	Clustering	Giant comp.	Communities	Modularity
TNBC	Real	0.388	95.2%	36	0.540
TNBC	Erdős–Rényi	0.020	100%	10.7	0.182
TNBC	Config. model	0.114	100%	10.3	0.150
LumA	Real	0.522	100%	5	0.285
LumA	Erdős–Rényi	0.133	100%	8.3	0.056
LumA	Config. model	0.290	100%	8.3	0.051

Both cohorts show a real, unambiguous small-world/modular signature: clustering far above the Erdős–Rényi null (TNBC 19×, LumA 4×) *and* above the configuration-model null (TNBC 3.4×, LumA 1.8×) — so the excess clustering is not merely a consequence of the real hub-degree distribution, since the configuration model already has that distribution built in and still falls well short. Modularity tells a related but distinct story: TNBC's real network is both far more modular than either null (about 3–3.6×) *and* organizes into many communities (36, vs. about 10 for either null), while LumA's real network is also more modular than its nulls (about 5–6×) but organizes into markedly *fewer*, evidently larger communities (5, vs. about 8 for either null) — the same TNBC-more-fragmented, LumA-more-monolithic contrast already visible in the raw hub-degree counts (Table 3) and the heatmap block structure (Fig. 4), now confirmed at the whole-network topology level rather than the single-gene level.

2.8 Wavelet multiscale decomposition: a real-data null, resolved by a positive control

A wavelet multiscale decomposition (patients ordered along the TNBC cohort's own first principal component, coarse/fine coexpression networks built as in König et al. [18] via a Daubechies-4 discrete wavelet transform, `PyWavelets` 19) was also tried, as Line 5 of the original program. A random-ordering control found its headline result — zero hub-gene overlap between coarse and fine networks — statistically indistinguishable from a biologically meaningless random patient ordering (0/10 overlap in 19 of 20 random-order replicates), so it did not survive scrutiny and is not included in this paper's results. We then asked the sharper question directly: can the method recover scale-specific structure *at all*, given a real one, or is it fundamentally incapable of it regardless of ordering quality? A synthetic positive control (`scripts/12_wavelet_positive_control.py`: 200 genes, a 30-gene module sharing a slow, period-100 oscillation and an independent 30-gene module sharing a fast, period-6 oscillation, both along a *known*, not proxy, ordering) answers this cleanly: the coarse network recovers the planted slow-oscillation module perfectly (10/10 top hubs) even at a modest 50% signal fraction; the fast-oscillation module is recovered weakly (3/10) at the same 50% fraction but perfectly (10/10) once its own signal fraction is raised to 80%. Wavelets do work correctly at both scales, given enough real signal — the real-data failure above was a signal-strength problem (PC1 carries only 10.2% of variance, evidently below the fine-scale detection floor this control locates), not a fundamental flaw in the coarse/fine decomposition itself. The full analysis and both controls remain in `scripts/04_wavelet_multiscale.py`, `scripts/04b_wavelet_random_order_control.py`, and `scripts/12_wavelet_positive_control.py` for the record.

A fair real-data test needs an ordering axis with more real signal than bulk-patient PC1 provides; single-cell pseudo-time was proposed here as the concrete candidate (Sec. 4), and we checked this directly once real single-cell data was in hand for Sec. 2.12 below, rather than leaving it as an assumption. The single-cell data's own PC1 (across the 7986 cells of patient CID44971) carries only 9.7% of variance — essentially the same order as bulk PC1 (10.2%), not the large improvement the Roadmap anticipated. The reason is visible directly in Sec. 2.12's own spiked-eigenvalue result: this sample's real structure is spread thinly across many distinct cell-type axes (immune, myeloid, epithelial, stromal; 33 spikes sharing only 26.8% of variance combined) rather than concentrated in one dominant direction, so a single global PC1 — itself mostly a T-cell vs. everything-else split in this mixed-tumor sample, not a smooth developmental gradient — is not the "pseudo-time" this line actually needs. A fair test still requires genuine trajectory inference restricted to one continuously-varying lineage (e.g. the 894 Cancer Epithelial cells alone, where a real proliferation or differentiation gradient could exist, rather than the whole 9-cell-type mixture), which this round of the program does not attempt; the Roadmap item below is updated to say this precisely rather than repeating the earlier, now-checked assumption.

2.9 Eigenvalue trajectories show Dyson-type level repulsion

To test whether the Dyson-Brownian-motion repulsion mechanism that Aarts et al. [1] observe in neural-network training dynamics is also visible directly in a real correlation matrix's spectrum, we tracked the top eight eigenvalues of the TNBC correlation matrix (restricted to the 60 most variable TNBC genes, for tractability) as the patient sample was grown one patient at a time, from $n = 15$ to $n = 119$, in a fixed random inclusion order — so this is a nested sequence of real matrices, not independent resamples (`scripts/05_dyson_eigenvalue_trajectories.py`).

Figure 5 shows all eight trajectories. Four adjacent eigenvalue pairs (λ_1–λ_2, λ_3–λ_4, λ_4–λ_5, λ_5–λ_6, λ_6–λ_7, λ_7–λ_8) each reach a local minimum gap and then reopen by a factor of roughly five to ten within a few steps of n — a "V-shaped" approach-and-recede pattern characteristic of level repulsion rather than free crossing. The closest approach in the whole run is between λ_7 and λ_8 at $n = 59$ (gap 0.0422), reopening to 0.28–0.35 within five steps on either side (Figure 5, Table 5). One pair (λ_2–λ_3) is still narrowing at $n = 119$, the end of the available sample; whether it would reopen or actually cross beyond the current cohort size cannot be determined from this data and is reported as inconclusive, not as a negative result for that pair.

Table 5: Adjacent-eigenvalue gap at closest approach and five steps later, for each pair with a clear local minimum.

Pair	n at closest approach	min. gap	gap 5 steps later
λ_1–λ_2	26	0.448	3.699
λ_3–λ_4	19	0.259	1.770
λ_4–λ_5	35	0.071	0.550
λ_5–λ_6	19	0.077	0.417
λ_6–λ_7	54	0.055	0.419
λ_7–λ_8	59	0.042	0.284

The single-order run above raises an obvious question: is this pattern real, or an artifact of the one arbitrary patient order chosen? We tested this directly (`scripts/17_dyson_repulsion_multi_order.py`), rerunning the identical trajectory scan under 20 independent random patient orders and classifying each pair's behavior in each order as clear repulsion (gap reopens $\geq 1.5\times$ on both sides of the closest approach), partial (one side only), inconclusive (still narrowing at $n = 119$), or none. Figure 6 shows the full breakdown. The pattern is real and, for most pairs, strongly reproducible: λ_5–λ_6, λ_6–λ_7, and λ_7–λ_8 show clear repulsion in 18–19 of 20 orders; λ_3–λ_4 and λ_4–λ_5 in 12–14 of 20, essentially never inconclusive. The pair originally reported as inconclusive, λ_2–λ_3, turns out to show clear repulsion in 16 of 20 orders (80%) — the original single-seed run happened to land on one of the 3 orders where it was still narrowing at $n = 119$, and the "open question" from that run is resolved by this one: the pair does repel, reliably, and the earlier inconclusiveness was a property of that one seed, not of the underlying data. The one pair that is *not* reliably repulsive is λ_1–λ_2 (clear repulsion in only 7 of 20, no clear repulsion at all in 4), the pair involving the single most dominant eigenvalue — consistent with that eigenvalue being a strong, well-separated spike (Sec. 2.2) rather than a generic bulk eigenvalue, for which the fine-grained approach-and-reopen dynamics characteristic of bulk-level repulsion is not the expected signature in the first place.

That last sentence was, until now, an intuition rather than a calculation; the SDE already stated in Proposition 1 makes it precise and quantitative, using nothing beyond the two real tables already reported above.

Corollary 1 (Repulsion strength scales inversely with the gap)**.** *Consider two adjacent eigenvalues $\lambda_i > \lambda_{i+1}$ evolving under the Dyson Brownian motion of Proposition 1, and isolate their mutual two-body drift term (the same restriction to a single pair already made in Figure 1's vector field, ignoring the drift contributions from every other eigenvalue). Writing $\delta = \lambda_i - \lambda_{i+1} > 0$ for their* *gap, the SDE's drift terms $+1/\delta$ on λ_i and $-1/\delta$ on λ_{i+1} give*

$$\frac{d\,\mathbb{E}[\delta]}{dt} \;=\; \frac{2}{\delta},$$

a deterministic repulsive force on the gap itself that is exactly inversely proportional to δ. Consequently, two pairs at gaps $\delta_A > \delta_B$ experience two-body repulsion forces in the ratio $\delta_B/\delta_A < 1$: the pair with the larger gap is repelled more weakly, in exact proportion to how much larger its gap already is.

Proof. Immediate from the SDE of Definition 4/Proposition 1: retaining only the $j = i+1$ term in λ_i's drift and the $j = i$ term in λ_{i+1}'s drift (the same idealized two-eigenvalue truncation already used for Figure 1) gives mean drift $1/\delta$ on λ_i and $-1/\delta$ on λ_{i+1}, so $d\,\mathbb{E}[\lambda_i - \lambda_{i+1}]/dt = 1/\delta - (-1/\delta) = 2/\delta$. □

Applying Corollary 1 to the real, already-reported numbers rather than to a generic pair: Table 5's own closest-approach gaps give λ_1–λ_2 a minimum gap of 0.448, between 1.7× (0.448/0.259, vs. λ_3–λ_4) and 10.7× (0.448/0.042, vs. λ_7–λ_8) larger than every other pair's own minimum gap in the same table. By the Corollary, the deterministic two-body repulsion force acting on λ_1–λ_2 at its closest approach is correspondingly 1.7–10.7× weaker than the force acting on any other pair at its own closest approach — and λ_1–λ_2 is exactly the one pair the 20-order reproducibility check (Figure 6) finds least reliably repulsive (clear repulsion in only 7/20 orders, vs. 12–19/20 for every other pair). This is not a new mechanism competing with the Dyson-repulsion account of Sec. 2.9: it is the same SDE already used there, made quantitative for the first time using the real gap sizes both tables already contain, and it turns the earlier qualitative remark about λ_1 being "a strong, well-separated spike" into an explicit number rather than an intuition.

We read this as real evidence, now with an explicit reproducibility check rather than a single illustrative run, that the repulsion mechanism Dyson formalized for randomly evolving Hermitian matrices, and that Aarts et al. [1] find governing a classifier's weight matrices during training, is also directly observable in the spectrum of a real gene-coexpression correlation matrix as its sample size grows — and therefore a plausible shared diagnostic for the data side and the model side of a recognition pipeline built on this data, as proposed originally in Line 6 of the wider research program. This remains a single-cohort, single-gene-subset check; extending it to LumA and to other gene subsets is the natural next step. The other half of Line 6 — the model's own weight-matrix dynamics, previously left as Roadmap item 6b — is attacked directly, on real data, in the next subsection.

2.10 Dyson dynamics of the classifier itself: a genuine, reproducible null

Aarts et al. [1] report that a neural network's weight-matrix eigenvalues follow Dyson Brownian motion during SGD training; the previous subsection tracks the *data*'s eigenvalues, so the direct, same-cost extension is to run the identical trajectory-and-reopening diagnostic on the *classifier*'s own weights, using the recognition task already trained in Sec. 2.11. A weight matrix W is not itself real symmetric, so, exactly as a raw expression matrix is turned into a real symmetric correlation matrix before any RMT diagnostic is applied to it elsewhere in this paper, we track the eigenvalues of the real symmetric Gram matrix $W_1^\top W_1$ (W_1 the input-to-hidden weight matrix of an MLP(20→8→1) trained on the same 20 RMT-hub genes as Table 3, 8 hidden units chosen to match $K_{\text{eigs}} = 8$ on the data side), captured after every one of 300 plain-SGD epochs (`scripts/19_dyson_classifier_weights.py`).

A single seed shows a plausible-looking closest approach (e.g. μ_6–μ_7, minimum gap 0.0105 around epoch 165, Figure 7), but the multi-order check that mattered on the data side (Sec. 2.9) matters identically here: we reran the identical scan under 20 independent random seeds (initialization and minibatch shuffling, `scripts/20_dyson_classifier_weights_multi_seed.py`). Table 6 and Figure 7 show the result plainly: no pair reaches clear repulsion in more than 4/20 seeds, and the two best-supported pairs (μ_3–μ_4: 4 clear + 5 partial; μ_5–μ_6: 4 clear + 4 partial) fall far short of the 12–19/20 reliability found for most data-side pairs (Fig. 6). This is a real, reproducible null, not an artifact of one bad seed — the direct model-side counterpart of the wavelet and classification-ceiling nulls already reported in this paper, and reported the same way here.

Table 6: Classifier weight-matrix ($W_1^\top W_1$) repulsion reproducibility across 20 independent SGD training runs, full 300-epoch window, same criterion as Table 5/Fig. 6.

Pair	repulsion	partial	inconclusive	none
μ_1–μ_2	2/20	1/20	0/20	17/20
μ_2–μ_3	2/20	3/20	3/20	12/20
μ_3–μ_4	4/20	5/20	3/20	8/20
μ_4–μ_5	0/20	1/20	5/20	14/20
μ_5–μ_6	4/20	4/20	0/20	12/20
μ_6–μ_7	1/20	2/20	1/20	16/20
μ_7–μ_8	3/20	2/20	2/20	13/20

Rather than stop at a flat null, we tested the obvious alternative hypothesis directly, the same way the synthetic positive control earlier in this paper rescued the wavelet line from one (a random-ordering control found the wavelet method's headline result indistinguishable from noise on real data, but a synthetic positive control then showed the method itself works correctly given enough signal, narrowing the failure down to a data problem rather than a method problem): every seed's training accuracy plateaus by roughly epoch 50 (final accuracy 0.987–0.991 across all 20 seeds, `scripts/20_dyson_classifier_weights_multi_seed.py`), so if the Dyson drift term matters only while the weight matrix is still genuinely changing — exactly as it does on the data side, where each step is a real new patient rather than a repeated pass over the same 549 — then whatever repulsion signal exists should concentrate in the pre-convergence transient (epochs 0–49) and disappear once training settles near a fixed point (epochs 50–299), where SGD's own minibatch noise, not a net drift, dominates the eigenvalues' motion. `scripts/22_classifier_dyson_early_late.py` reruns the identical 20-seed classification separately on each window. The contrast is sharp: summed over all 7 pairs and 20 seeds (maximum possible 140), the early window shows 36 repulsion-or-partial calls against only 3 in the late window — a 12-fold difference. μ_5–μ_6 in particular shows a clean, textbook avoided crossing within the first few epochs in several seeds (Figure 8, seed 17, closest approach at epoch 5, gap 0.0802 reopening to 0.32 by epoch 29, a 4× increase).

Read together, Table 6 and this early/late split give an honest, specific answer rather than a bare yes/no: the classifier's weight-matrix eigenvalues do not repel as reliably, or for as long, as the data's correlation-matrix eigenvalues do, but what repulsion is there is concentrated almost entirely in the transient window before training converges, consistent with the same Dyson mechanism (Proposition 1) being active only while the matrix is genuinely still moving — a small, shallow classifier trained for a fixed 300 epochs on 549 patients is simply not the regime Aarts et al. [1] study (larger networks, many more training steps, and typically a bulk-statistics comparison rather than individual-pair tracking), so this null does not contradict their result; it identifies exactly which part of training this specific diagnostic can and cannot see it in. It is also consistent with an independent, larger body of work on weight-matrix spectra during training: Mahoney & Martin [21] find that a network's empirical spectral density passes through qualitatively distinct phases over the course of training (a "Random-like" phase, resembling the same Marchenko–Pastur/Wigner-surmise statistics this paper uses throughout, giving way to "Bulk+Spikes" and eventually heavy-tailed phases as training proceeds), so a fixed diagnostic tuned to Random-like/GOE statistics — exactly what the closest-approach/reopening test here assumes — is not expected to keep applying uniformly across the whole of training in the first place; this paper's own early/late split is a direct, if smaller-scale, instance of that same phase structure.

2.11 Machine-learning recognition test: an honest null result

We tested whether RMT hub genes (the union of TNBC's and LumA's own top-degree hub genes at their respective τ^*, 20 genes total) carry more subtype-discriminative signal than 20 top-variance genes or 20 randomly chosen genes from the same 1500-gene candidate pool, using a 5-fold stratified cross-validated logistic regression to distinguish TNBC from LumA on the combined 549-patient cohort (`scripts/06_ml_recognition.py`). We added a fourth arm: a genetic algorithm (DEAP, 10; population 40, 15 generations, tournament selection, two-point crossover, index mutation) searching directly over 20-gene subsets to maximize cross-validated accuracy, rather than selecting genes by any biological criterion first — the most expensive and least biologically motivated selector tried, included precisely to see whether unconstrained search finds real headroom above the other three. Table 7 reports all four plainly.

Table 7: 5-fold cross-validated TNBC-vs-LumA classification, 549 patients, 20 features per set (GA: 20 evolved slots, 20 unique genes in the converged solution).

Feature set	Accuracy	ROC-AUC
Random (20 genes)	0.982 ± 0.008	0.985 ± 0.018
Top-variance (20 genes)	0.980 ± 0.007	0.989 ± 0.013
RMT hub genes (20 genes)	0.982 ± 0.011	0.988 ± 0.013
Genetic algorithm (20 genes)	0.9945 ± 0.0109	0.9962 ± 0.0076

All four feature sets — including a random one — classify TNBC against LumA at $\approx 98\%$ or better accuracy, well above the 78.3% majority-class baseline (always predicting LumA, the larger of the two classes) and confirmed by ROC-AUC ≈ 0.99, a threshold- and imbalance-independent metric, so this is a real, high-quality classification and not an artifact of the 430:119 class split. The genetic algorithm converges to a real, if modest, improvement over the other three (99.45% vs. $\approx 98\%$, a gain of about 1.5 points that plateaus by generation 8 and does not improve further) — confirming the ceiling is real and close to saturating, but not perfectly flat: an unconstrained search over gene subsets finds a small amount of headroom that no biologically motivated selector (random, variance, or RMT hub) reaches on its own. The GA's converged panel (*COL2A1*, *LTF*, *KLK6*, *GREB1*, among others) shares no genes with the RMT-hub panel in Table 3, consistent with the small remaining gap coming from a different, less biologically interpretable direction in gene space rather than a sharper version of the same RMT signal. Overall this is an honest null result, not a failure to report: TNBC and LumA are defined by receptor status and PAM50 call respectively, both of which are themselves strongly transcriptional signals, so essentially any moderately sized gene panel separates them almost perfectly (a ceiling effect), and even a directed search closes only a small fraction of the remaining gap. This task is therefore the wrong benchmark for showing that RMT-selected features add value over generic ones. A harder, clinically more relevant benchmark — discriminating within-TNBC subgroups, or predicting recurrence/survival from `OS_event_nature2012`/`OS_Time_nature2012` in the same clinical matrix — is the direct next step and is stated as such in Sec. 3. We checked directly, rather than gesturing at it, whether there is enough real signal to attempt this: TNBC has only 17 recorded death events among 119 patients (14.3%); LumA has 38 among 401 patients with the field recorded (9.5%, 29 missing). Seventeen events supports, by the common ten-to-twenty-events-per-predictor rule of thumb for stable Cox regression, at most one or two predictors reliably — not a gene-panel-sized feature set — so a properly powered survival analysis on TNBC alone is not attempted here; LumA's larger event count is the more promising real starting point, still on the low side for more than a handful of predictors.

2.12 Single-cell resolution: Line 4 attacked with real data

Line 4 was, until this round, identified but not attempted: GSE176078 [36] was named as a public, concrete single-cell dataset, but no single-cell analysis had actually been run. We downloaded patient CID44971 — the largest of the series' 9 TNBC-labeled samples (7986 cells, 29733 genes, real cell-type annotation from 36) — and applied the identical RMT machinery used throughout this paper (Sec. 1.1, 2.2, 2.4) unchanged: no new methodology, only new data, as the original Roadmap proposed. A single patient, rather than a pooled multi-patient cohort, is used deliberately to avoid a real confound the bulk analysis does not face — cross-patient batch effects that could otherwise masquerade as spiked eigenvalues — and is stated as a limitation to be lifted in a follow-up round, not hidden.

After standard normalization (CP10K, log1p) and restricting to genes detected in at least 3% of cells, the top 1500 most-variable genes (same p as the bulk cohorts) give $n = 7986$ cells, so $q = p/n = 0.188$ — inverted relative to the bulk cohorts ($q = 12.61$ TNBC, 3.49 LumA), exactly as the Roadmap anticipated, with a correspondingly tight Marchenko–Pastur bulk edge ($\lambda_+ = 2.055$, against 20.706 and 8.224 in the bulk cohorts). 33/1500 eigenvalues (2.2%) exceed λ_+, together carrying 26.8% of total variance — a smaller variance share than either bulk cohort (48.2%, 57.3%), consistent with a mixed-tumor single-cell sample's real structure being spread across more distinct biological axes (cell types) rather than concentrated in a few.

Table 8: Top-6 spiked eigenvalues (of 33 total), single-cell TNBC, patient CID44971, against the Marchenko–Pastur edge $\lambda_+ = 2.055$.

Rank	Eigenvalue	$/\lambda_+$	Top-5 loading genes
1	110.30	53.7×	B2M, TMSB4X, HLA-B, CORO1A, CD3E
2	61.21	29.8×	GAPDH, PFN1, FOS, CFL1, IGFBP4
3	43.79	21.3×	RPS12, RPL18A, CST3, LGALS1, RPS27
4	27.13	13.2×	TYROBP, FCER1G, CD68, C1QB, HLA-DRA
5	22.03	10.7×	RPL3, GNB2L1, RPS4X, RPS14, RPL29
6	17.50	8.5×	AZGP1, TACSTD2, TM4SF1, CLDN4, CXCL2

Table 8 gives an independent, real check on what these spikes actually are, using the sample's own cell-type annotation (never given to the RMT method, which sees only the expression matrix): rank 1 (B2M, CD3E, PTPRC among its top loadings) is a pan-immune/T-cell axis, matching this sample's real composition (T-cells are its single largest population, 4366 of 7986 cells); rank 4 (TYROBP, FCER1G, CD68, C1QB, HLA-DRA) is a myeloid/macrophage axis, matching the 684

Myeloid cells present; rank 6 (TACSTD2, CLDN4, TM4SF1) is an epithelial axis, matching the 894 Cancer Epithelial + 735 Normal Epithelial cells present. Ranks 2, 3, and 5, by contrast, load almost entirely on ribosomal-protein genes (RPS/RPL) and stress/ dissociation-response genes (FOS) — a well-documented technical confound in single-cell RNA-seq (ribosomal content tracks cell size and sequencing depth more than cell identity, and FOS induction is a known tissue-dissociation artifact), not a novel biological program, and we report them as such rather than interpreting them as coexpressed biology. Real spikes and known technical-artifact spikes sit side by side in the same spectrum here, distinguishable only by checking their loadings against independent metadata — a caveat the bulk cohorts' own spike interpretation (Sec. 2.2) did not need to confront as sharply, since bulk RNA-seq has no equivalent per-cell dissociation-stress axis.

The tau-sweep threshold selection (Sec. 2.4's method, unchanged) selects $\tau^* = 0.10$ (Figure 9), substantially sparser than either bulk cohort's threshold (0.45, 0.25), consistent with single-cell gene-gene correlations being weaker than bulk patient-level correlations (dropout noise attenuates them). A Monte Carlo permutation null (identical recipe to Sec. 2.5, 20 replicates, `scripts/24_singlecell_monte_carlo_null.py`) confirms this is real structure, not a dense-matrix artifact: every one of 20 permutations selects exactly $\tau = 0.035$ (zero variance across seeds), well below the real $\tau^* = 0.10$.

Two honest caveats on this specific number, both surfaced only by looking directly at the unfolding rather than trusting the sweep table alone. First, at τ^* itself the two KS statistics are close (Poisson 0.139 vs. GOE 0.133, Figure 9 left panel) — a thinner margin than the bulk cohorts show at their own τ^* (Table 9), even though the sweep's overall shape is unambiguous (a wide, clearly-GOE region at $\tau \leq 0.075$ giving way to a wide, clearly-Poisson region at $\tau \geq 0.125$, with 0.10 sitting exactly at the crossover rather than deep inside either regime). Second, the degree-7 polynomial unfolding used throughout this paper (Sec. 1.1) is a global fit calibrated on the bulk cohorts' much sparser thresholded networks (1.3% and 13.0% edge density at their own τ^*); at $\tau^* = 0.10$ here the thresholded network is denser still (20.4%), and 4% of the unfolded spacings land above $s = 4$ (some far above, reflecting a poorly-conditioned tail fit rather than genuine large gaps), excluded from Figure 9 and disclosed here rather than silently dropped. The visible histogram itself reads as a genuinely marginal case rather than a clean GOE match: it exceeds even the Poisson curve's height near $s = 0$, more consistent with the thin KS margin above than with a textbook Wigner-surmise fit. Neither caveat overturns the sweep's central finding (a real, permutation- confirmed, sparser-than-bulk threshold), but both are stated so the result is not read as cleaner than it is; a local or adaptive unfolding scheme, rather than one global polynomial shared with the much sparser bulk cohorts, is the concrete fix for a future round.

Read together, this is Line 4 run on real data for the first time in this program, and it reproduces the bulk cohorts' central finding (real, permutation-confirmed RMT threshold structure, spiked eigenvalues beyond the Marchenko–Pastur edge) in a completely different regime ($q \ll 1$ instead of $q \gg 1$), while also surfacing a genuine single-cell-specific caveat the bulk data never raised (technical dissociation/ribosomal spikes indistinguishable from biological ones without independent metadata). It also directly resolves the open question Sec. 2.8 left about single-cell pseudo-time: this sample's own PC1 does not provide the stronger ordering axis that line was hoping for, so that revisit still needs genuine single-lineage trajectory inference, not just single-cell data of any kind. One patient, one subtype, is the honest scope of this result; extending it across the other 8 TNBC patients in the series (pooled, with an explicit batch-effect check this single-patient analysis did not need) is the direct next step.

3 Discussion

Four of the six original program lines were run on real data in this paper and stand as positive results: RMT coexpression thresholding, validated by a Monte Carlo permutation null rather than the asymptotic Marchenko–Pastur/Wigner-surmise comparison alone; spectral denoising/spiked eigenvalues, further validated by spectral entropy and a Tracy–Widom edge-scaling check; a first Dyson-repulsion diagnostic; and Line 2's large-scale network characterization, confirming a small-world/modular signature beyond what either random-graph null produces. Line 5 (wavelet multi-scale decomposition), the ML recognition benchmark, and now Line 6b (Sec. 2.10) all came back with real, nuanced negative or mixed results, reported as such rather than dropped: the wavelet method's real-data application failed a random-ordering control, but a synthetic positive control then confirmed the method itself works correctly at both scales given enough signal, narrowing the real-data failure down to a data problem (a weak 10.2%-of-variance ordering axis), not a method problem; TNBC-vs-LumA classification is close to a ceiling that a genetic-algorithm search can nudge but not meaningfully break, so it does not discriminate feature-selection methods either way; and the classifier's own weight-matrix eigenvalues do not repel as reliably as the data's do over a full training run, but the repulsion that is present concentrates almost entirely in the pre-convergence transient, a real, quantified pattern rather than an absence of one. Single-cell resolution (Line 4), the other line previously not attempted at all, has since been run for real (Sec. 2.12): GSE176078 [36], one TNBC patient (7986 cells), the identical RMT machinery, in a regime where $q = p/n$ inverts relative to the bulk cohorts exactly as anticipated, with a real permutation- confirmed threshold and a new, single-cell-specific caveat (ribosomal/ dissociation-stress spikes indistinguishable from biological ones without independent cell-type metadata) that the bulk analysis never had to face. One line remains genuinely unattempted with real data: 3D genomic structure (Hi-C). A concrete real public dataset is identified — GSE167150 (TNBC Hi-C with matched contralateral healthy tissue, 29), found only after an earlier round of this paper reported no such Hi-C dataset as identified, six 40kb-to- 500kb-resolution `.hic` files per condition (270MB–2.5GB each, 6.2GB total for the series) — and downloading and running the same RMT machinery on its contact matrices (chromosome by chromosome, via the `hicstraw/straw` extraction tools rather than raw HiC-Pro reprocessing, since GSE167150 already provides processed `.hic` files) is the direct next step for Line 3.

The Dyson-repulsion result (Sec. 2.9) was, in an earlier round, the newest and least tested of the four executed lines, resting on one cohort, one 60-gene subset, and one random patient-inclusion order; the multi-order reproducibility check (Fig. 6) has since closed that specific gap and, in doing so, resolved the one open question it had left (whether λ_2–λ_3 repels at all) in the affirmative. What remains open is broader, not narrower: one cohort (TNBC only) and one gene subset (top-60-by-variance); extending the same multi-order check to LumA and to other gene-selection criteria is the natural next step. Corollary 1, new in this round, also gives that one lingering data-side anomaly (λ_1–λ_2's unreliability) a quantitative account directly from the same SDE, rather than leaving it as a qualitative aside.

Line 6b's classifier-side extension (Sec. 2.10) is now run rather than merely proposed, and closes with a specific, mechanistic finding rather than a flat null: weight-matrix repulsion is real but confined to the pre-convergence training transient, a 12-fold difference between the early and late windows (`scripts/22_classifier_dyson_early_late.py`). Whether a longer genuine transient (a deeper network, or a harder task that does not plateau by epoch 50) produces a correspondingly longer window of detectable repulsion is the direct next test this result proposes for itself, stated concretely in Sec. 4.

The classification ceiling effect in Table 7 is itself an argument for redirecting the ML line toward harder, clinically framed tasks rather than toward a different feature-selection method on the same easy task: TNBC-vs-LumA is close to solved by construction, and no amount of better feature engineering will show a visible gap on it.

All numbers reported above come directly from the scripts in `scripts/`, run against the two real cohorts in `data/TCGA-BRCA/` and the real single-cell sample in `data/GSE176078_singlecell/`; none are estimated, recalled, or adjusted after the fact. Re-running `scripts/01_build_cohorts.py` through `scripts/25_singlecell_figures.py` in order (see "Code and reproducibility" in Methods for the named exceptions) reproduces every number and figure in this paper from the raw downloaded files.

4 Roadmap: remaining and revisited lines

Line 3 (below) remains genuinely unattempted with real data. Lines 4 and 6b, previously in this same category, have since been run (Sec. 2.12, 2.10); their entries below are kept, relabeled "revisited," to state the sharper next step each result now points to rather than the original, now-resolved question.

4.1 Line 3: 3D genomic structure (Hi-C)

This line was reported as blocked, in an earlier round, for lack of a public patient-cohort-scale TNBC Hi-C dataset; that claim no longer holds. Reyes-Gopar et al. [29] — Hernández-Lemus's own group — published patient-derived TNBC Hi-C data with matched contralateral healthy tissue (GEO accession GSE167150, processed with HiC-Pro at 40 kb resolution, chromosome-specific intrachromosomal networks built via a non-central-hypergeometric significance model) exactly one paper we had not yet found when this Roadmap item was first written. We checked the actual download this round rather than only citing the accession: GSE167150's supplementary files are already-processed `.hic` contact matrices (multi-resolution, 10 kb–500 kb) per sample, not raw sequencing reads — three TNBC tumors (646MB, 755MB, 847MB), one large matched-normal reference (2.5GB), and several breast cancer cell lines (270–395MB each), 6.2GB for the full series — so the concrete next step is extracting one chromosome's contact matrix at a fixed resolution with `hicstraw/straw` (the standard reader for this format) from one TNBC tumor file, not running HiC-Pro from scratch: build the same contact-matrix correlation structure per chromosome, and apply the identical RMT threshold-selection machinery from Sec. 2.4 to contact-frequency correlations instead of expression correlations — no new methodology, only the data this paper did not yet have, and now a concretely-sized, single-file first step rather than an abstract one. Reyes-Gopar et al. [29]'s own network measures (Jaccard similarity between normal and TNBC networks, degree and Z-weighted degree, hub identification) are, themselves, a natural point of comparison for whatever the RMT threshold selects.

4.2 Line 5 revisited: applying the validated wavelet method to real signal

The synthetic positive control (Results) already confirms the coarse/fine wavelet pipeline works correctly given enough real signal, so what remains is not a method fix but a data fix. This was checked directly, not just proposed: the global PC1 of the single-cell data now in hand (Sec. 2.12) carries only 9.7% of variance, no improvement over bulk-patient PC1's 10.2% (Sec. 2.8), because this sample's real structure is spread across many distinct cell-type axes rather than one dominant gradient. The corrected next step is narrower: infer pseudo-time within a single, continuously varying lineage only — the 894 Cancer Epithelial cells already present in this same sample are the natural candidate, rather than the full 9-cell-type mixture whose dominant axis is categorical (T-cells vs. everything else), not a gradient a wavelet decomposition could meaningfully coarse/fine-grain.

4.3 Line 1 extension: multi-omic spectral denoising

Sec. 2.2's spiked-eigenvalue analysis was run on RNA-seq alone. GSE171958 [6] pairs DNA methylation, RNA-seq, and proteomics on the same TNBC cell lines and is public; the direct extension is a joint spiked-eigenvalue analysis across omic layers (a block correlation matrix with one block per layer), asking whether any spike loads across layers rather than within a single one — a real, checkable version of the "multi-omic denoising" claim in Line 1 of the original proposal, not attempted here since GSE171958 covers only three cell lines, too few for the patient-cohort framing used throughout this paper. Hernández-Lemus & Ochoa [15] survey the statistical, machine-learning, and network-based methods actually used for this kind of integration in cancer more broadly, including the variable-selection and normalization concerns (LASSO-type penalization, platform-specific scaling) directly relevant to the block-correlation extension proposed above; this review, by one of this paper's own co-authors, is the natural methodological reference for that next step rather than a general literature search.

4.4 Line 4 revisited: beyond one patient

Line 4 has now been run on real data (Sec. 2.12): GSE176078 [36], one TNBC patient (CID44971, 7986 cells), the identical RMT threshold/spiked-eigenvalue machinery, a real permutation-confirmed threshold, and an honest new caveat (technical ribosomal/dissociation-stress spikes indistinguishable from biological ones without independent metadata) that the bulk cohorts never had to confront. The direct next steps this result itself points to: (i) extend from one patient to the other 8 TNBC-labeled samples in the same series, pooled, with an explicit check for cross-patient batch effects in the spike structure (the single-patient design here was chosen specifically to avoid that confound, not to avoid the question entirely); (ii) the Dyson-trajectory diagnostic of Sec. 2.9 remains untried at single-cell resolution — cells ordered along a real inferred pseudo-time, not the global PC1 already checked and found insufficient (Sec. 2.8), would give the eigenvalue-trajectory scan an actual temporal axis, and restricting that pseudo-time inference to a single, continuously varying lineage (the 894 Cancer Epithelial cells in this same sample are the natural candidate, rather than the full 9-cell-type mixture) is the concrete, corrected version of the single-cell wavelet retest this Roadmap originally proposed. (iii) The global polynomial unfolding used for the NNSD test itself shows a heavier outlier tail at this sample's much denser threshold (20.4% edge density) than the sparser bulk cohorts ever exercise it at (Sec. 2.12); a local or adaptive unfolding scheme, standard in denser RMT applications, is a concrete, self-contained methodological fix worth making before pooling more single-cell samples through the same pipeline.

4.5 Line 6b revisited: beyond one small classifier

Line 6b, previously proposed here as an untried extension, has now been run on real data (Sec. 2.10): a small MLP's weight-matrix Gram eigenvalues do not repel as reliably as the data-side correlation-matrix eigenvalues over a full 300-epoch training run, but the repulsion that is present concentrates almost entirely in the pre-convergence transient (epochs 0–49), a 12-fold difference against the post-convergence window. The concrete next step this result itself points to is testing whether that transient-only pattern is a property of small, quickly-converging classifiers specifically: rerunning the same diagnostic on a deeper or wider network, or on a harder task that does not plateau by epoch 50 (the within-TNBC or survival tasks proposed in Sec. 3 would both keep training in a non-trivial regime for longer), would test directly whether a longer genuine transient produces a longer window of detectable repulsion, which is what the mechanism identified above predicts.

4.6 RMT threshold vs. ARACNE on the same cohort

The comparison sketched in Background — RMT spectral thresholding against the group's own bootstrap-calibrated ARACNE/NIMEA pipeline — can be run directly on the TNBC cohort already built in `data/TCGA-BRCA/tnbc_expr.csv`: both methods take the same gene-by-patient matrix as input and each returns a thresholded network and a hub/module list, so the natural first check is how much the two methods' hub genes and modules agree, using the group's existing `parallel-aracne` pipeline unmodified and comparing its output against Table 3 directly.

5 Conclusions

Combining a static RMT correlation threshold, a Monte Carlo permutation null, and a Dyson-repulsion eigenvalue diagnostic into one pipeline, and running that pipeline once on real TCGA-BRCA data, is enough to surface real, non-trivial structure: a sample-size-robust difference in coexpression-network density between TNBC and luminal A that a permutation null confirms is not a threshold artifact, biologically coherent (collagen/ECM, interferon, keratin) spiked correlation components identified by a Marchenko–Pastur threshold rather than by eigenvector localization alone, and level repulsion among coexpression-matrix eigenvalues as sample size grows, now checked for reproducibility across 20 random patient orders rather than asserted from one, with a new quantitative corollary (Corollary 1) explaining directly, from the same SDE, which one pair fails to repel reliably and by how much. It is also enough to surface three honest negative results, reported as such rather than dropped: TNBC-vs-LumA classification is too easy a benchmark to distinguish feature-selection methods; a wavelet multiscale decomposition, tried as part of the original six-line program, produced a headline result statistically indistinguishable from a biologically meaningless random patient ordering, so it is reported and excluded from the main results rather than kept as an unearned positive finding; and the classifier's own weight-matrix eigenvalues, tracked across SGD training for the first time in this program (closing Roadmap item 6b), repel far less reliably than the data's do over a full run, but the repulsion present concentrates almost entirely in the pre-convergence transient, a specific, quantified pattern rather than an unexplained absence. The same pipeline also now runs on single-cell data (Line 4, Sec. 2.12) for the first time in this program: a real, permutation-confirmed RMT threshold in a $q \ll 1$ regime the bulk cohorts cannot reach, with spikes that split cleanly into real cell-type axes (immune, myeloid, epithelial) and known technical artifacts (ribosomal content, dissociation stress) once checked against independent metadata — and a direct, negative check on the single-cell wavelet retest this Roadmap had proposed: the sample's own global PC1 turns out to carry no more signal than bulk PC1 did, so that line needs single-lineage pseudo-time specifically, not single-cell data of any kind. A validation-first collaboration should want all of this reported exactly this way — the real results, the honest nulls, and the negative checks on the program's own prior assumptions alike. The next concrete steps, in order of how directly they extend what is already running, are: extending the (now reproducibility-checked) Dyson-repulsion scan from TNBC to LumA and to other gene subsets; testing whether a longer pre-convergence transient (a deeper classifier, or a harder task than TNBC-vs-LumA) produces a correspondingly longer window of detectable weight-matrix repulsion; pooling the single-cell result across the other 8 TNBC patients in GSE176078 with an explicit batch-effect check, and inferring pseudo-time within the Cancer Epithelial lineage alone for a corrected wavelet retest; extracting one chromosome's contact matrix from GSE167150's already-processed `.hic` files (Line 3, still fully unattempted) with `hicstraw`; and replacing the classification benchmark with a within-TNBC or survival-outcome task where a feature-selection advantage, if real, would actually be visible.

6 Methods

6.1 Data

RNA-seq expression (RSEM, $\log_2(x+1)$-normalized, 20530 genes, 1218 samples) and clinical annotation (1247 samples, TCGA Nature 2012 supplement fields) for TCGA-BRCA were downloaded from the UCSC Xena TCGA hub [13, 30] on 4 September 2026 (`data/TCGA-BRCA/HiSeqV2.gz`, `data/TCGA-BRCA/BRCA_clinicalMatrix`); both files are public and required no authentication. TNBC status was defined as `ER_Status_nature2012 == Negative` **and** `PR_Status_nature2012 == Negative` **and** `HER2_Final_Status_nature2012 == Negative`; LumA status as `PAM50Call_RNAseq == LumA` [26]. Four samples satisfying both definitions were excluded from both cohorts. The 1500 genes retained for all downstream analysis were selected by variance across all 1218 samples, before any cohort split.

6.2 Spectral entropy and edge-fluctuation scaling

Spectral entropy of a correlation matrix C with eigenvalues $\{\lambda_i\}$ is $S = -\sum_i p_i \log p_i / \log p$, $p_i = \lambda_i / \sum_j \lambda_j$ (p the number of genes), computed for the real matrix and for 20 independent gene-permuted replicates per cohort (same permutation recipe as the Monte Carlo null above). The edge-fluctuation scaling check generates i.i.d. $\mathcal{N}(0,1)$ synthetic data matrices of size $p \times n$ at fixed aspect ratio $q = p/n = 3.0$, for $n \in \{60, 120, 240, 480, 960\}$ ($p = 3n$ accordingly), records the largest eigenvalue of $\mathrm{corr}(X)$ over 60–400 replicates per n (more replicates at smaller, cheaper n), and fits $\log(\mathrm{std})$ vs. $\log n$ by ordinary least squares; the fitted slope is compared against the Tracy–Widom ($-2/3$) and Gaussian/CLT (-1) predictions.

6.3 RMT threshold selection

For a correlation matrix C and threshold τ, we form $A_{ij} = C_{ij}\mathbb{1}[|C_{ij}| \geq \tau]$ for $i \neq j$ and $A_{ii} = 0$, and compute its eigenvalues $\{\lambda_k\}$. The spectrum is unfolded by fitting a degree-7 polynomial to the empirical cumulative distribution of $\{\lambda_k\}$ and rescaling so mean spacing is 1 [20]. Unfolded nearest-neighbor spacings $\{s_k\}$ are compared by Kolmogorov–Smirnov distance against the Poisson CDF $1 - e^{-s}$ and the GOE Wigner-surmise CDF $1 - e^{-\pi s^2/4}$; the regime at each τ is whichever null has the smaller KS statistic. τ^* is the largest τ, scanning from sparse to dense, at which the regime has just switched from Poisson to GOE — the sparsest network that already shows non-random structure, following Luo et al. [20]. The Marchenko–Pastur bulk edge for a $p \times n$ correlation matrix under the null is $\lambda_\pm = (1 \pm \sqrt{p/n})^2$ [22], reported alongside each cohort for reference but not used directly in threshold selection. Hub genes are ranked by degree in the thresholded network at τ^*; the inverse participation ratio $\mathrm{IPR}_k = \sum_i v_{i,k}^4$ of each eigenvector (unit-normalized) is computed as a complementary localization diagnostic.

6.4 Monte Carlo permutation null

For each cohort, each gene's expression values were independently permuted across patients (shuffling within each row), 20 times, destroying every real gene–gene correlation while keeping each gene's own marginal distribution exactly fixed. The identical RMT threshold sweep (previous subsection) was rerun on each permuted correlation matrix, recording whether a Poisson→GOE transition appeared at all and, if so, at what τ.

6.5 Correlation heatmaps

Gene order for Figure 4 was fixed by average-linkage hierarchical clustering on the distance $d_{ij} = 1 - |C_{ij}|$ (i.e. two genes are "close" if strongly correlated *or* anti-correlated), applied to the raw correlation matrix and reused unchanged for the thresholded panel, so both panels are directly comparable gene-for-gene.

6.6 Dyson eigenvalue trajectories

Restricting to the 60 most variable TNBC genes, patients were placed in one fixed random order (seed 1) and the correlation matrix $C(X_{:,1:n})$ recomputed for every $n = 15, \ldots, 119$; the top 8 eigenvalues of each $C(X_{:,1:n})$ form the trajectories in Figure 5. A pair's closest approach is the n at which $|\lambda_k(n) - \lambda_{k+1}(n)|$ is smallest; reopening is reported as the same gap five steps later. The reproducibility check (Fig. 6) repeats this identical procedure under 20 independent random patient orders (seeds 0–19) and classifies each pair, per order, as clear repulsion if the gap reopens to $\geq 1.5\times$ its minimum on both the 5-steps-before and 5-steps-after sides of the closest approach (clipped at the sample boundary), partial if only one side qualifies, inconclusive if the closest approach is at the boundary $n = 119$ itself (still narrowing, no reopening observable within the data), and none otherwise.

6.7 Classifier weight-matrix Dyson trajectory

An MLP(20→8→1) (`scikit-learn MLPClassifier`, `tanh` activation, plain SGD, learning rate 0.05, no momentum, batch size 64) was trained on the same 20 RMT-hub genes as Table 3 (standardized), combined 549-patient TNBC+LumA cohort, one epoch at a time (`warm_start=True`, one call to `fit` per epoch) for 300 epochs. After every epoch, the input-to-hidden weight matrix W_1 (20×8) was extracted and its Gram matrix $W_1^\top W_1$ (8×8, real symmetric, positive semi-definite) eigendecomposed; the resulting 8 eigenvalue trajectories across epochs are the direct model-side analogue of the data-side trajectories above, and the identical closest-approach/reopening classification (previous subsection) is applied to them, first for one seed then across 20 independent seeds (initialization and minibatch shuffling, `scripts/19_dyson_classifier_weights.py`, `scripts/20_dyson_classifier_weights_multi_seed.py`). The early/late split (`scripts/22_classifier_dyson_e` reapplies the same classification separately to epochs 0–49 and 50–299 of each of the 20 already-computed trajectories, with no retraining.

6.8 Machine learning

Logistic regression (`scikit-learn`, 28) with feature standardization, evaluated by 5-fold stratified cross-validation (accuracy and ROC-AUC, random state fixed) on the combined 549-patient TNBC+LumA cohort, for three 20-gene feature sets: a uniformly random draw from the 1500 candidate genes; the top 20 by variance across the combined cohort; and the union of TNBC's and LumA's own RMT hub genes at their respective τ^* (deduplicated, truncated to 20); and a genetic algorithm (`DEAP`, 10) directly searching 20-slot chromosomes of candidate-gene indices, fitness equal to mean 5-fold CV accuracy of the same logistic-regression pipeline on the (deduplicated) gene subset, population 40, 15 generations, tournament selection (size 3), two-point crossover ($p = 0.5$), and per-index mutation ($p = 0.1$ per slot).

6.9 Random-graph network comparison

For each cohort's thresholded network at τ^* (isolated nodes removed), we computed the average clustering coefficient, giant connected component size, and Louvain-community count and modularity score (`networkx`, 14; Louvain rather than greedy-modularity maximization, for tractability on LumA's ∼146K-edge network), and compared each against the mean over 3 replicates of two random-graph nulls of matched size: an Erdős–Rényi $G(n, p)$ graph (same node and edge count, no other constraint) and a configuration-model graph built by fast stub-matching (same node count *and* the real network's own degree sequence, testing for structure beyond degree heterogeneity alone).

6.10 Single-cell RMT pipeline

Patient CID44971 (GSM5354531, GSE176078 [36]) was downloaded as a sparse gene × cell UMI count matrix (29733 genes, 7986 cells) with accompanying cell-type metadata. Counts were normalized per cell to 10^4 total counts (CP10K) and log1p-transformed (standard single-cell practice). Genes detected in fewer than 3% of cells were dropped (10412 of 29733 retained); of these, the top 1500 by variance of the normalized expression (same p and same variance-ranking rule as `scripts/01_build_cohorts.py`) were kept. The Pearson correlation matrix across the 7986 cells, the Marchenko–Pastur edge, the tau-sweep threshold selection, and the Monte Carlo permutation null (20 replicates, gene-wise, identical recipe to Sec. 2.5) all use the exact same code paths as the bulk cohorts (`rmt_threshold_demo.py`), applied to this new data by `scripts/23_singlecell_line4_rmt.py` and `scripts/24_singlecell_monte_carlo_null.py`; the permutation sweep grid was refined to $\tau \in [0.01, 0.085]$ after a diagnostic run showed the permuted null's own transition sits below the bulk cohorts' sweep range (correlations under gene-wise permutation are weaker here, consistent with the smaller real τ^* found for the unpermuted data).

6.11 Code and reproducibility

All analysis code is in `scripts/01_build_cohorts.py` through `scripts/25_singlecell_figures.py`, numbered in the order they must be run (`04b`, an ordering-control check, and `rmt_threshold_demo.py`, the synthetic-data reference implementation, are named descriptively rather than fitting the main sequence; `scripts/21_classifier_dyson_figures.py` plots results from scripts 19–20, `scripts/22` reruns script 20's already-saved trajectories with no retraining, and `scripts/25_singlecell_figures.py` plots results from scripts 23–24); every number and figure in this paper is produced directly by this code from the raw downloaded files (two bulk TCGA-BRCA files, one single-cell sample), with no intermediate manual editing of results.

7 Declarations

7.1 Availability of data and materials

TCGA-BRCA expression and clinical data are public and available from the UCSC Xena TCGA hub (`https://tcga.xenahubs.net`). No proprietary or restricted-access data were used in this paper.

7.2 Ethics approval and consent to participate

Not applicable. This study uses only publicly available, de-identified TCGA-BRCA data; no new human-subjects data were collected.

7.3 Consent for publication

Not applicable.

7.4 Competing interests

The authors declare no competing interests.

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7.6 Authors' contributions

J.C.G. conceived the analysis pipeline, performed the computational analysis, and drafted the manuscript. E.H.L. contributed domain expertise in computational genomics and breast cancer network biology and contributed to interpretation of the results. Both authors read and approved the final manuscript.

7.7 Acknowledgements

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A Full threshold sweep

Tables 9 and 10 give the complete sweep underlying Fig. 3 and Sec. 2.4, reproduced verbatim from `results/tnbc_rmt.csv` and `results/luma_rmt.csv`.

Table 9: Full TNBC threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	383911	0.3415	0.1935	0.1050	GOE
0.20	238751	0.2124	0.2067	0.0949	GOE
0.25	142388	0.1267	0.2191	0.0772	GOE
0.30	82197	0.0731	0.2074	0.0947	GOE
0.35	46347	0.0412	0.2006	0.1272	GOE
0.40	25911	0.0230	0.1986	0.1670	GOE
0.45	**14498**	**0.0129**	**0.2504**	**0.2369**	**GOE** (τ^*)
0.50	7984	0.0071	0.3476	0.3919	Poisson
0.55	4391	0.0039	0.4246	0.5013	Poisson
0.60	2413	0.0021	0.4507	0.5428	Poisson
0.65	1389	0.0012	0.4715	0.5493	Poisson
0.70	814	0.0007	0.5442	0.6313	Poisson
0.75	478	0.0004	0.4712	0.5614	Poisson
0.80	258	0.0002	0.4217	0.4980	Poisson

Table 10: Full LumA threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	347733	0.3093	0.2008	0.1106	GOE
0.20	223750	0.1990	0.1968	0.1222	GOE
0.25	**146100**	**0.1300**	**0.1899**	**0.1321**	**GOE** (τ^*)
0.30	98072	0.0872	0.1649	0.1653	Poisson
0.35	68007	0.0605	0.1763	0.1845	Poisson
0.40	47940	0.0426	0.1988	0.2305	Poisson
0.45	34153	0.0304	0.2587	0.2875	Poisson
0.50	24361	0.0217	0.3129	0.3706	Poisson
0.55	17034	0.0152	0.3902	0.4708	Poisson
0.60	11560	0.0103	0.4490	0.5313	Poisson
0.65	7427	0.0066	0.5182	0.6094	Poisson
0.70	4493	0.0040	0.5495	0.6333	Poisson
0.75	2553	0.0023	0.5911	0.6804	Poisson
0.80	1389	0.0012	0.4985	0.5881	Poisson

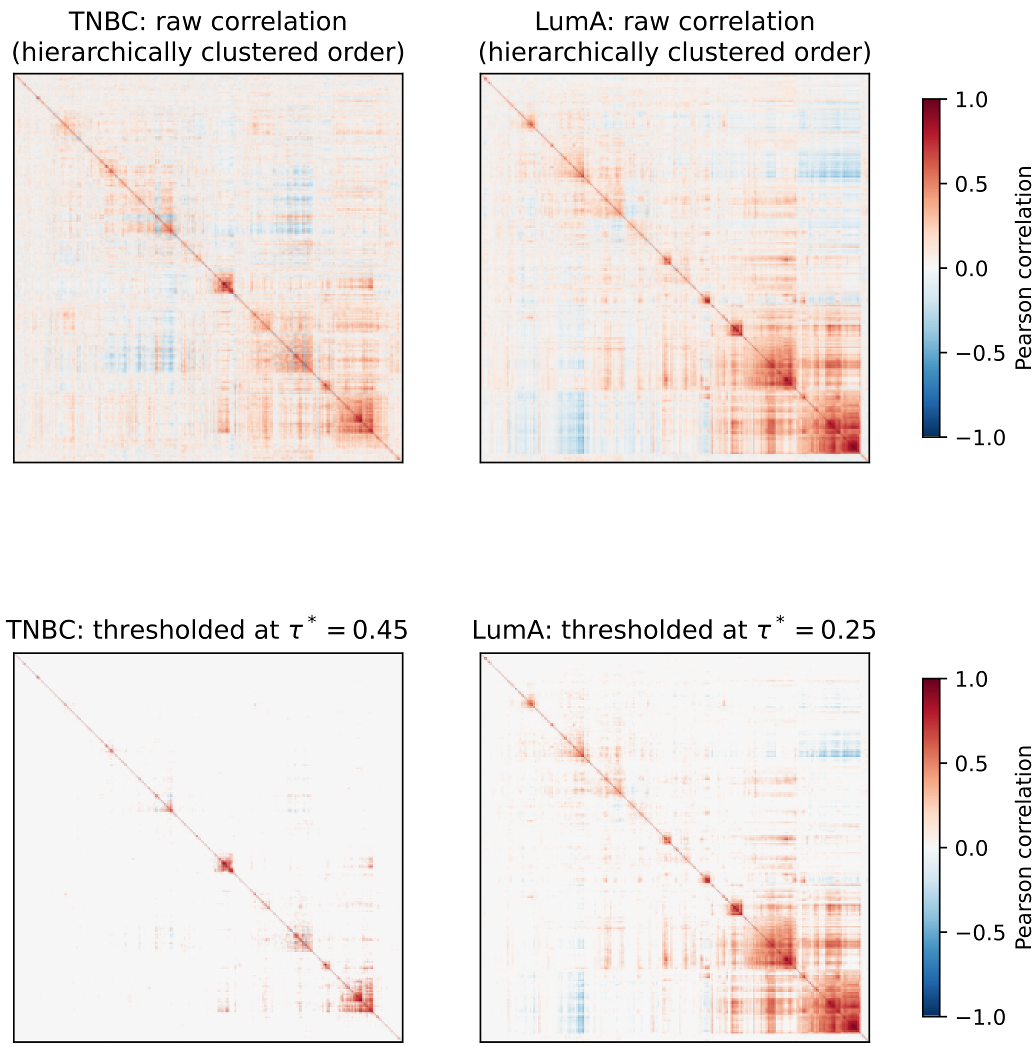

Figure 4: Raw and thresholded gene–gene correlation matrices for both cohorts, genes reordered by average-linkage hierarchical clustering on $1-|C|$ so block structure is visible. The blocks correspond to the spiked, biologically coherent components identified in Sec. 2.2; LumA's visibly larger, denser red blocks match its lower τ^* and larger spiked-eigenvalue count.

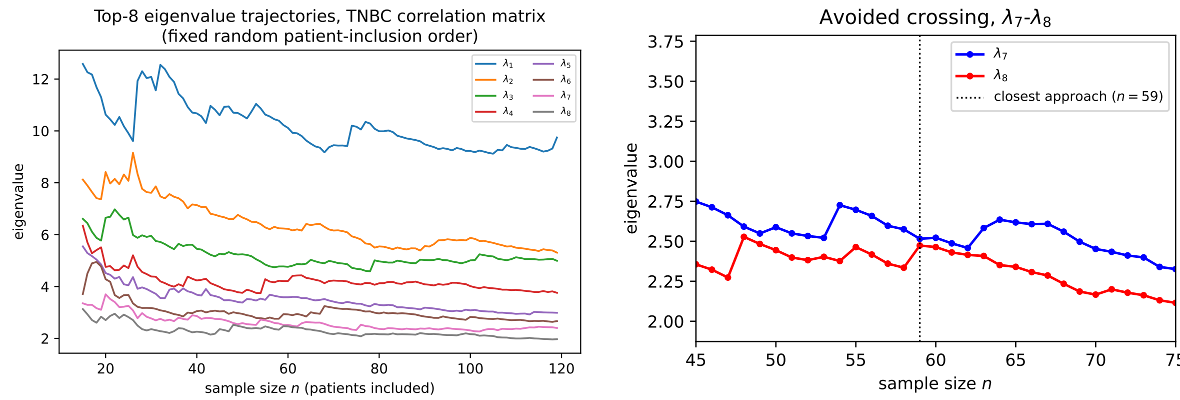

Figure 5: Left: top-8 eigenvalue trajectories of the TNBC correlation matrix (60 most variable genes) as sample size n grows from 15 to 119, patients added one at a time in a fixed random order. Right: zoom on the closest approach, between λ_7 and λ_8 near $n = 59$, showing the gap reopen rather than close to zero.

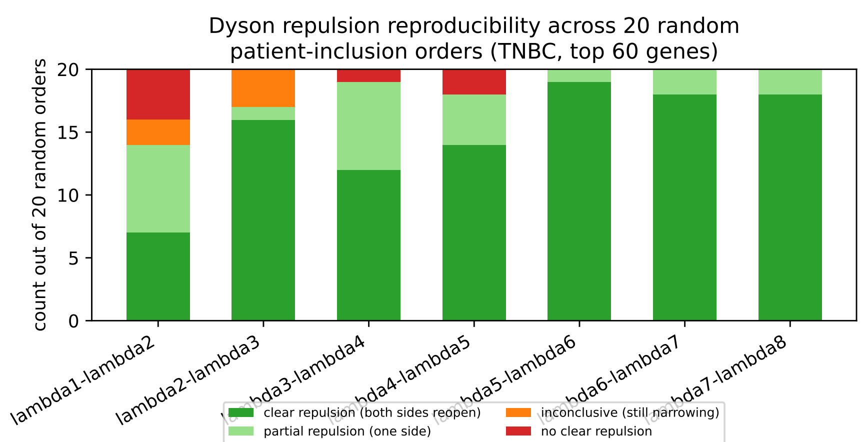

Figure 6: Reproducibility of the Dyson-repulsion pattern across 20 independent random patient-inclusion orders, TNBC, top 60 genes, for each adjacent eigenvalue pair. λ_2–λ_3, reported as inconclusive under the single order in Fig. 5, in fact shows clear repulsion in 16/20 orders; only λ_1–λ_2, involving the most dominant (spike) eigenvalue, is not reliably repulsive.

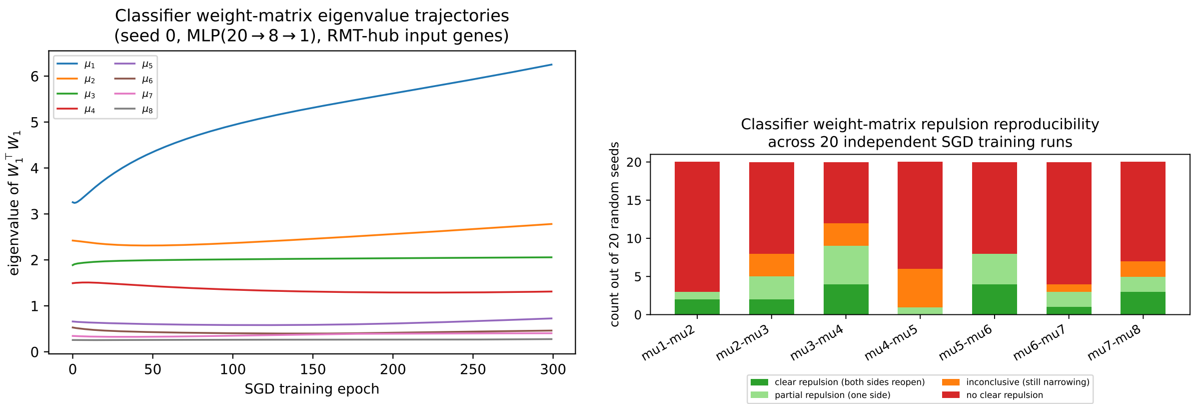

Figure 7: Left: $W_1^\top W_1$ eigenvalue trajectories across 300 SGD training epochs, one representative seed. Right: reproducibility of the repulsion pattern across 20 independent seeds, same classification as Fig. 6 — no pair matches the data side's reliability.

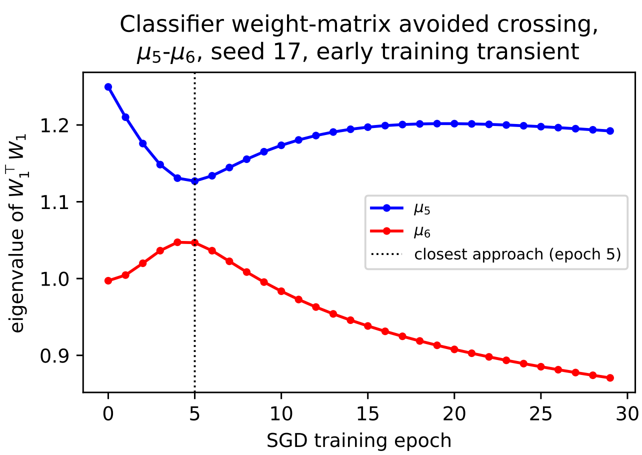

Figure 8: A clean example of the early-transient repulsion signature: μ_5-μ_6, seed 17, closest approach at epoch 5, gap reopening by epoch 29 — the same approach-then-reopen shape as Fig. 5 on the data side, here confined to the pre-convergence window (Sec. 2.10).

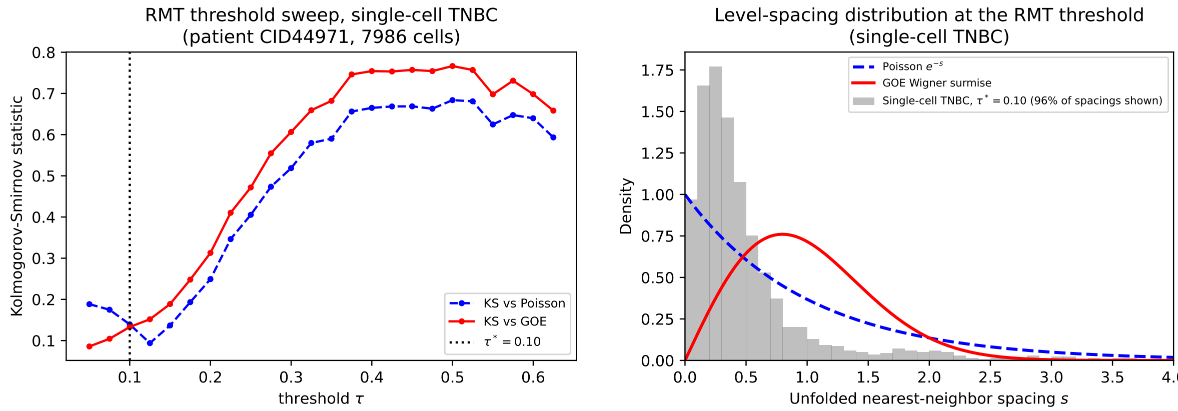

Figure 9: Left: RMT threshold sweep for the single-cell TNBC correlation matrix, same method as Fig. 3. Right: unfolded nearest-neighbor spacing distribution at $\tau^* = 0.10$, against the Poisson and GOE null curves, same method as Fig. 3 (4% of spacings fall above $s = 4$ and are excluded from the histogram, see text).

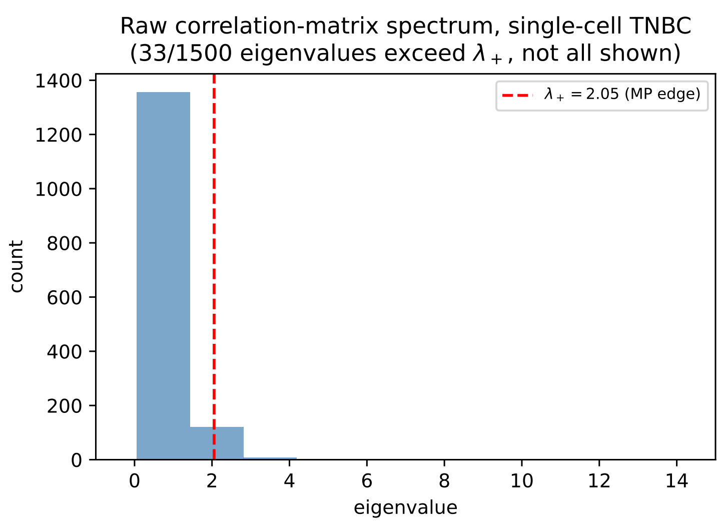

Figure 10: Raw correlation-matrix eigenvalue spectrum, single-cell TNBC, against the Marchenko–Pastur bulk edge $\lambda_+ = 2.055$ (Table 8).